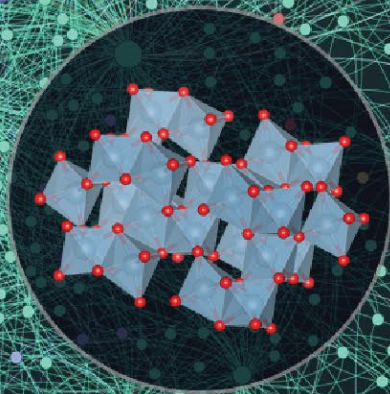
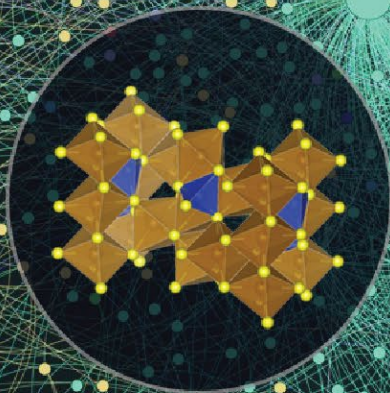


Gaussian processes

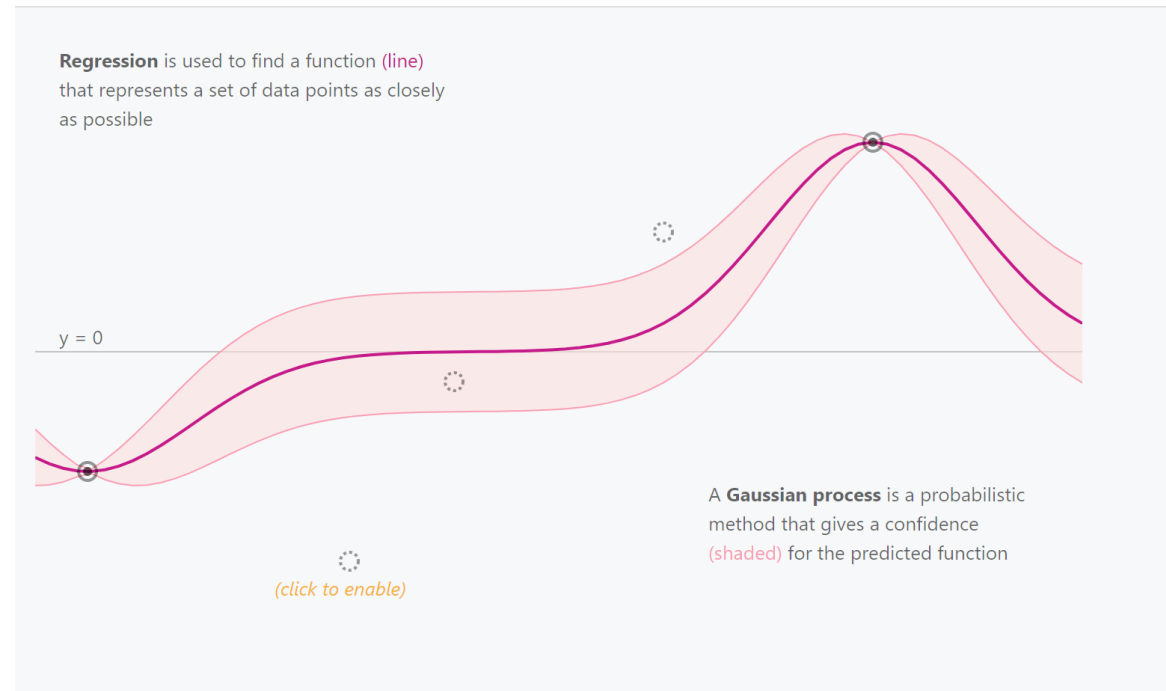




Gaussian process can be tricky to understand!

A Visual Exploration of Gaussian Processes

How to turn a collection of small building blocks into a versatile tool for solving regression problems.



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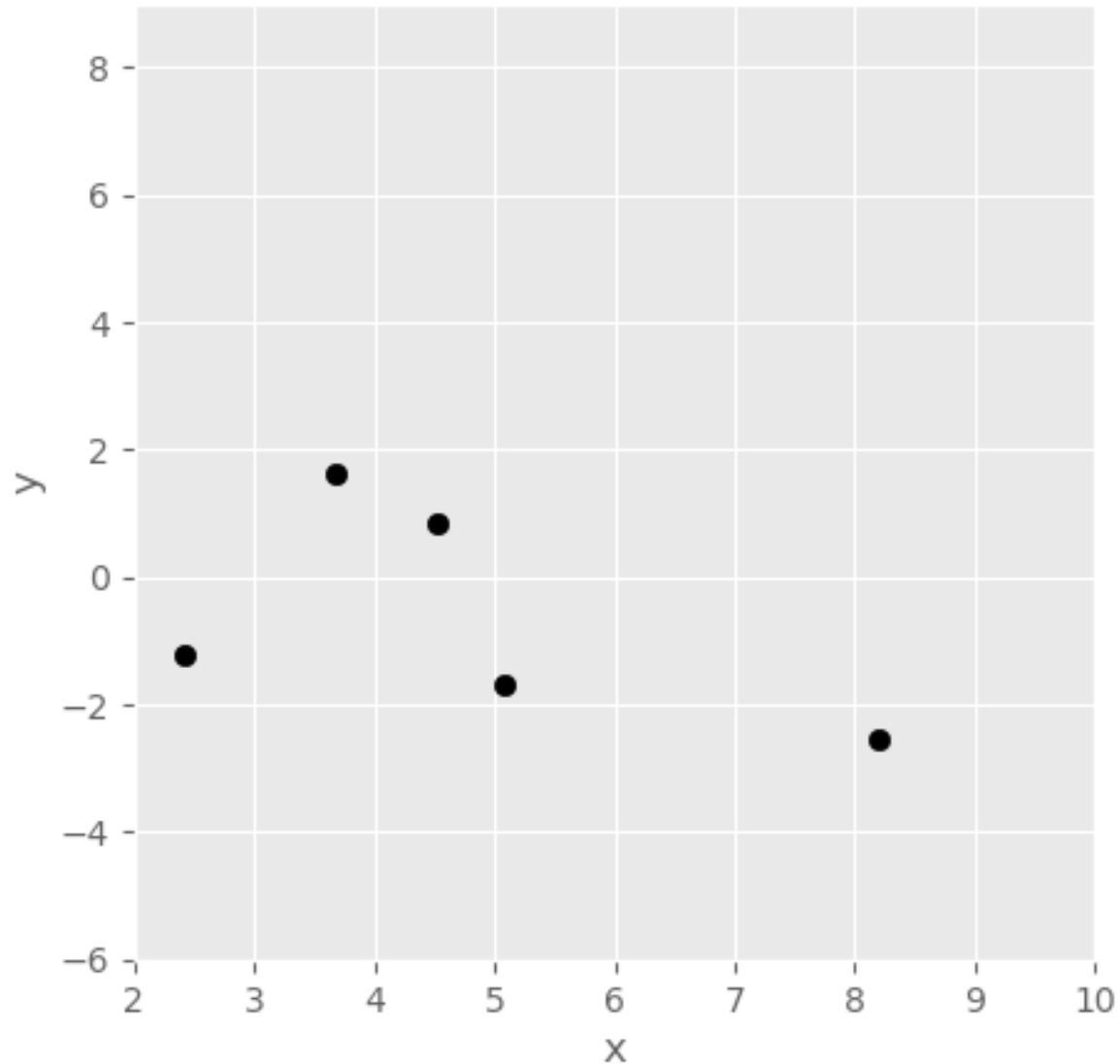
April 2, 2019

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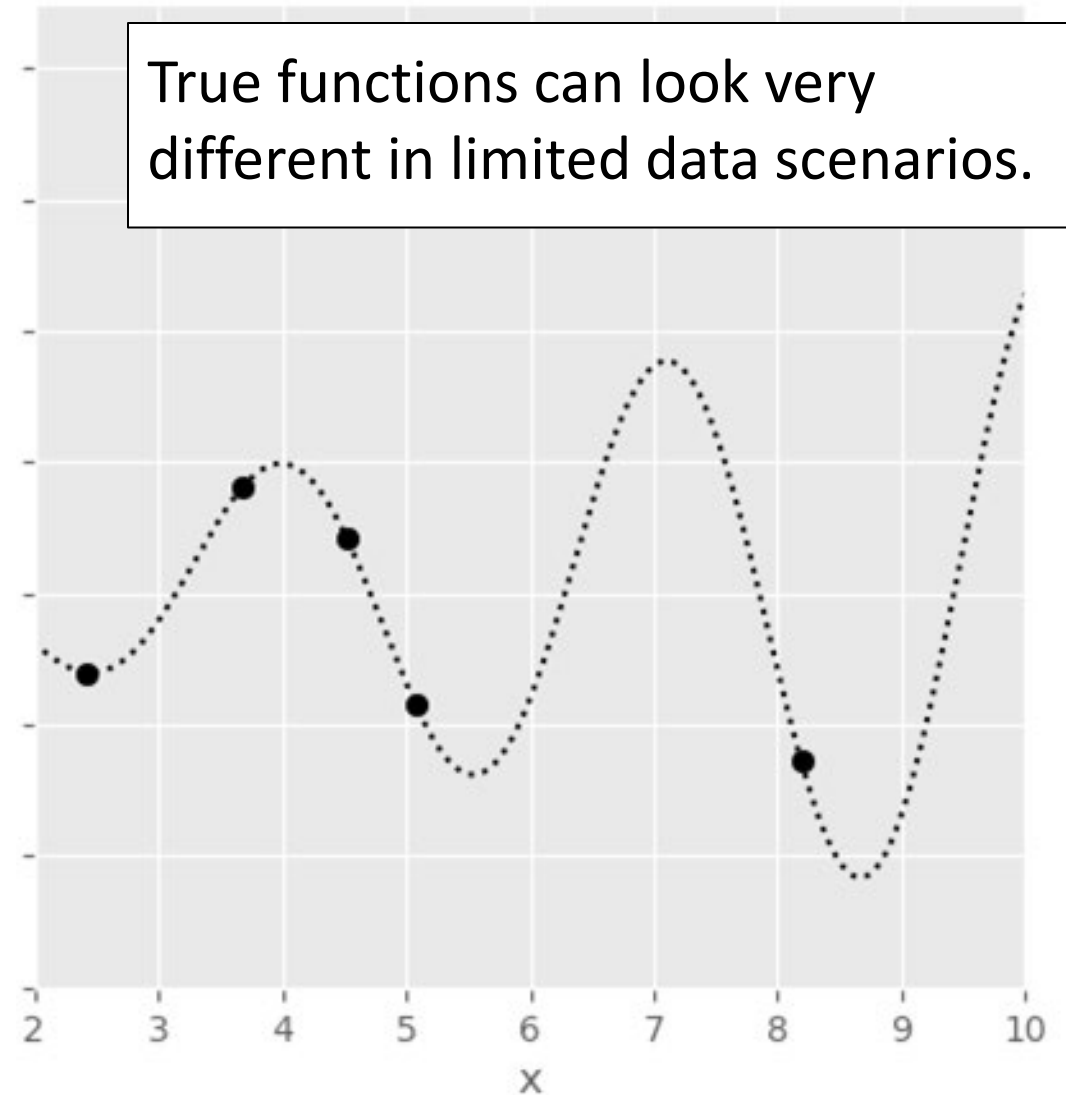
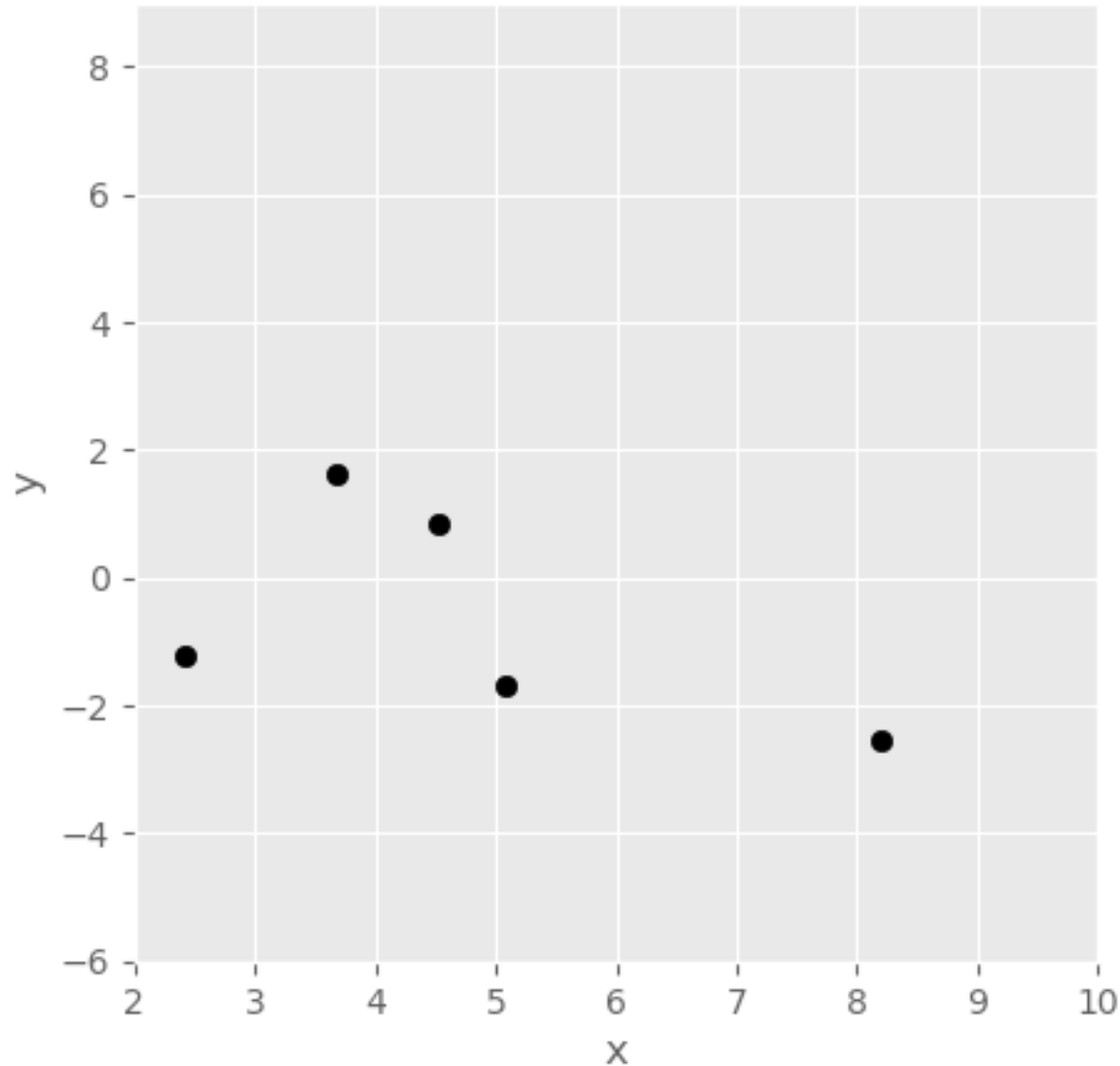
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Why should we care about this?

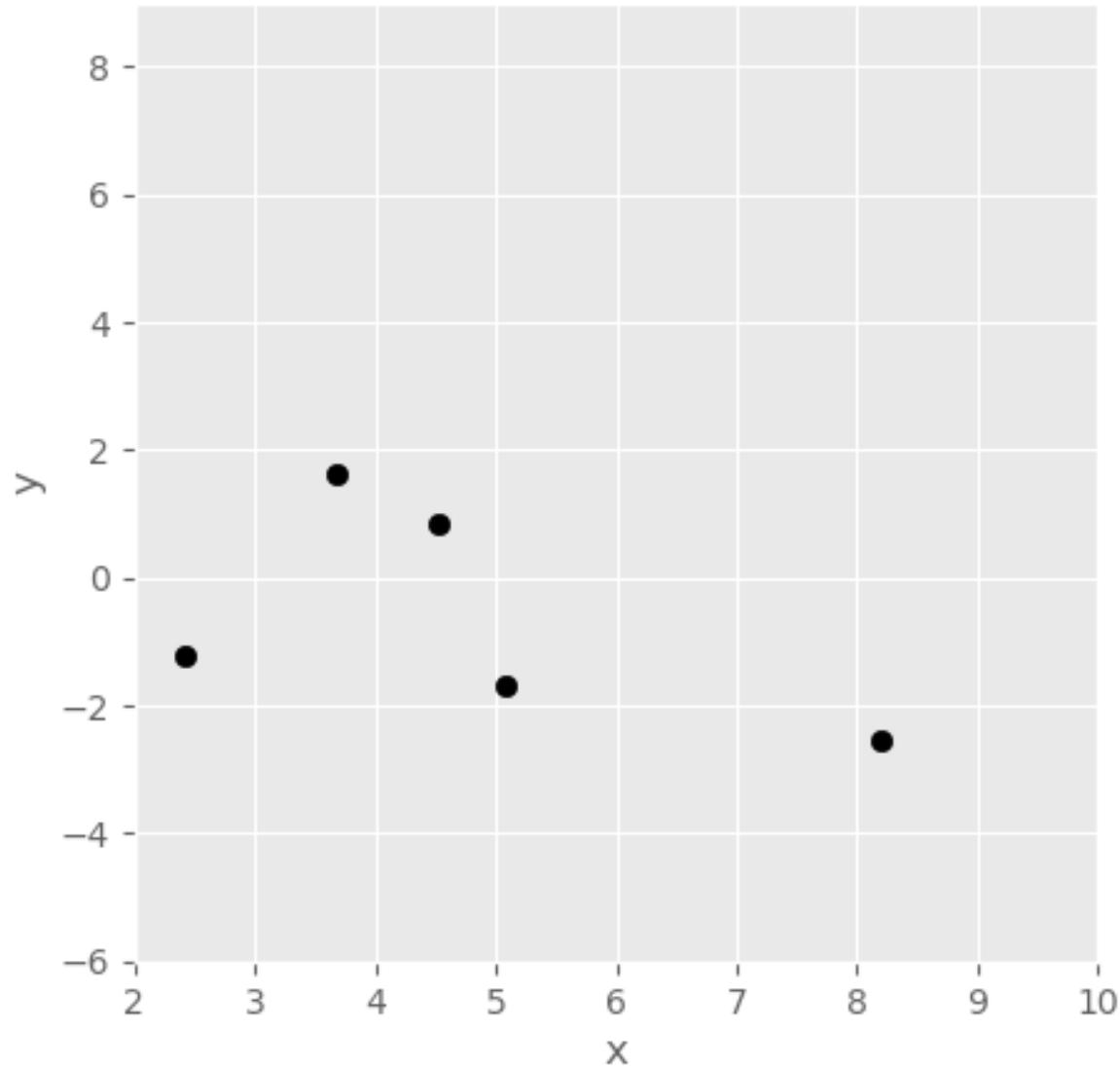
How would you fit this function?



Why should we care about this?



Why should we care about this?





Naïve bayes had us multiplying prior by likelihoods for predictions

$$p(y|X) = p(y) * p(y|X_1) * p(y|X_2) \dots$$



The function could take infinite forms

$$C(y|X) = g(X)$$

Consider linear regression

$$C(y|X) = g(X, \beta) = \beta_0 + X_1\beta_1 + X_2\beta_2 + \dots$$

Assumptions:

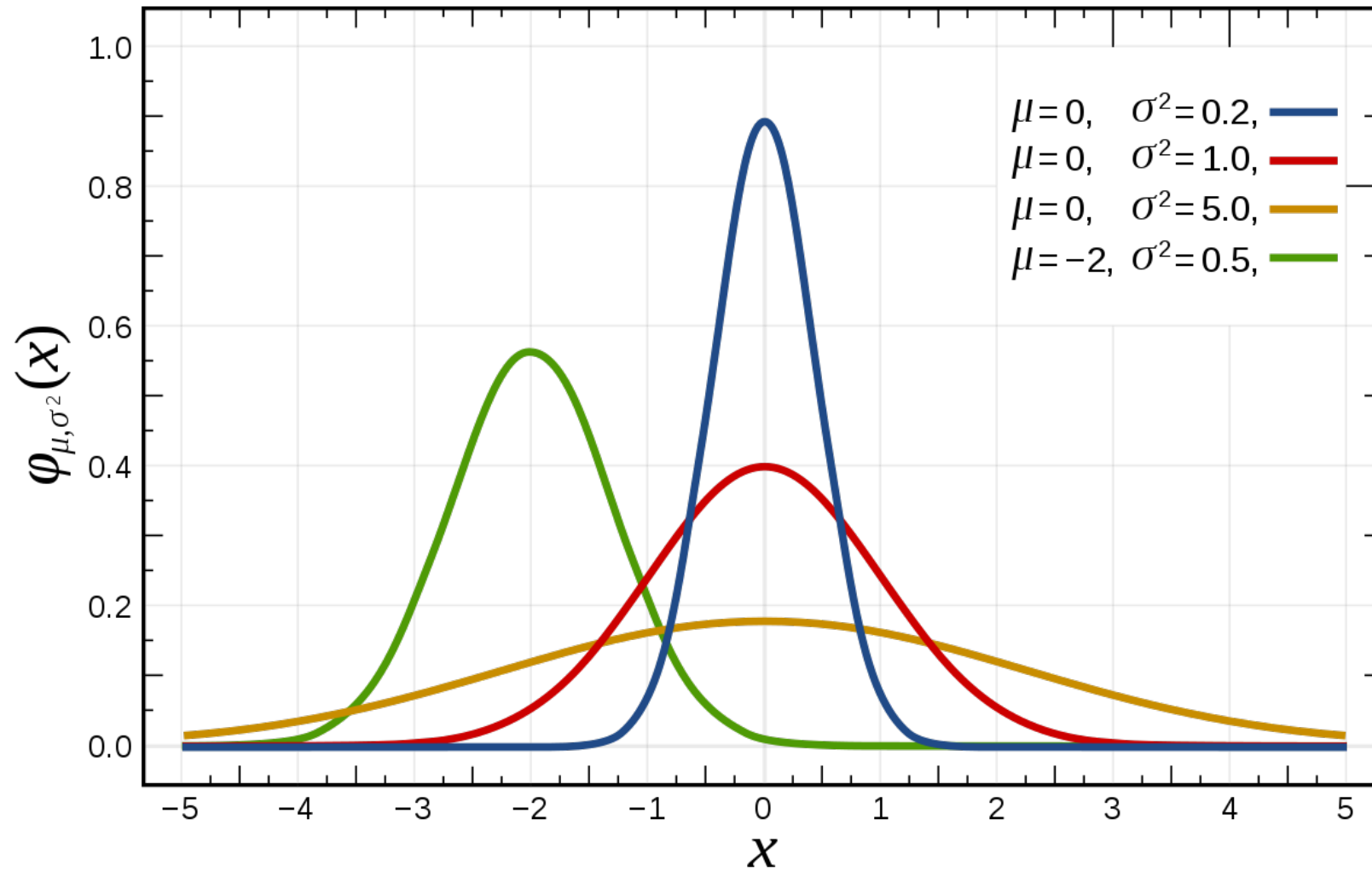
Parametric (data is normally distributed) and linear relationships between variables

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$
$$\epsilon_i \sim N(0, \sigma)$$

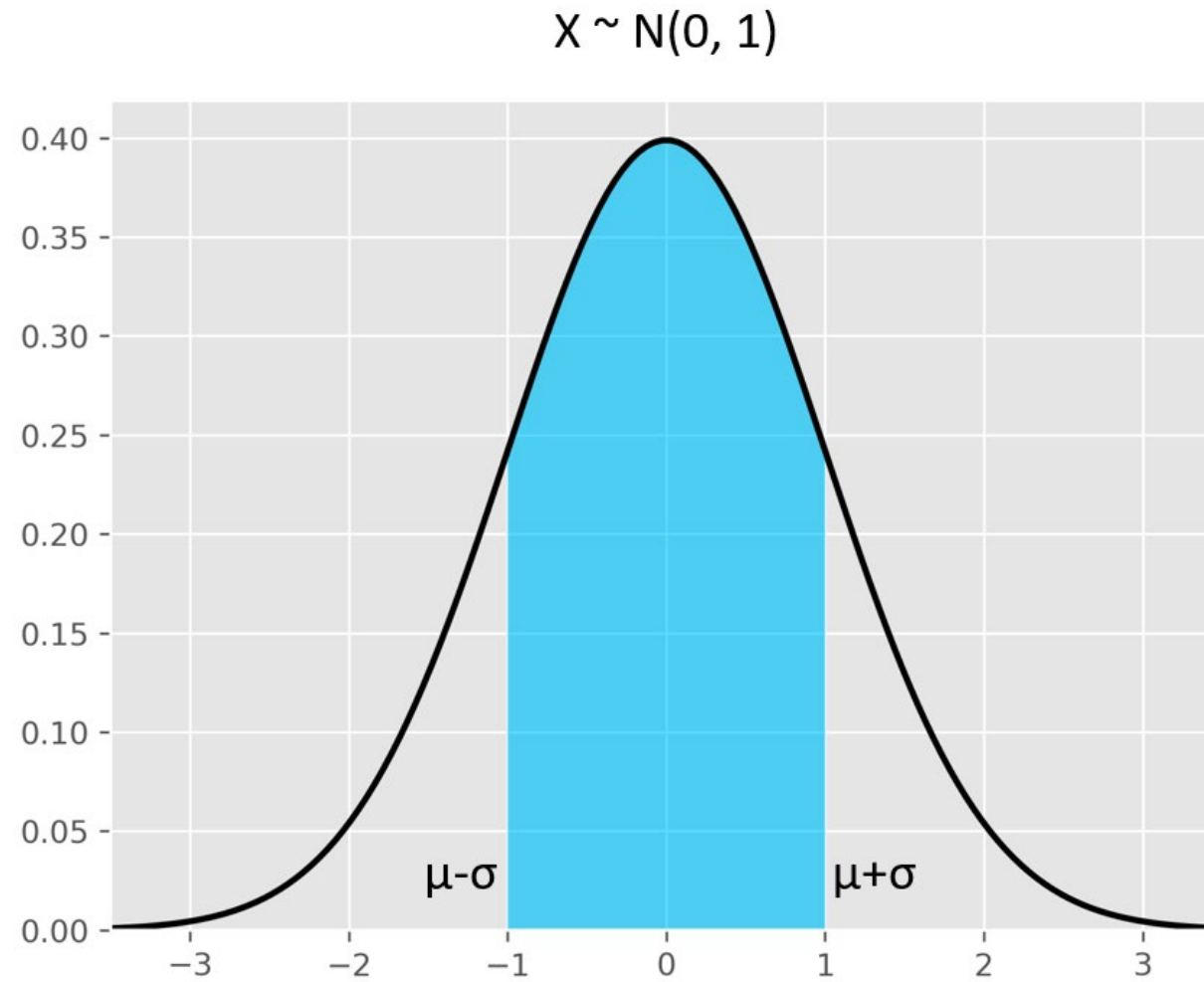
Logistic regression also possible

$$C(y|X) = \frac{1}{1 + \exp(\beta X)}$$

Instead of linear or non-linear options... let's use Gaussians!

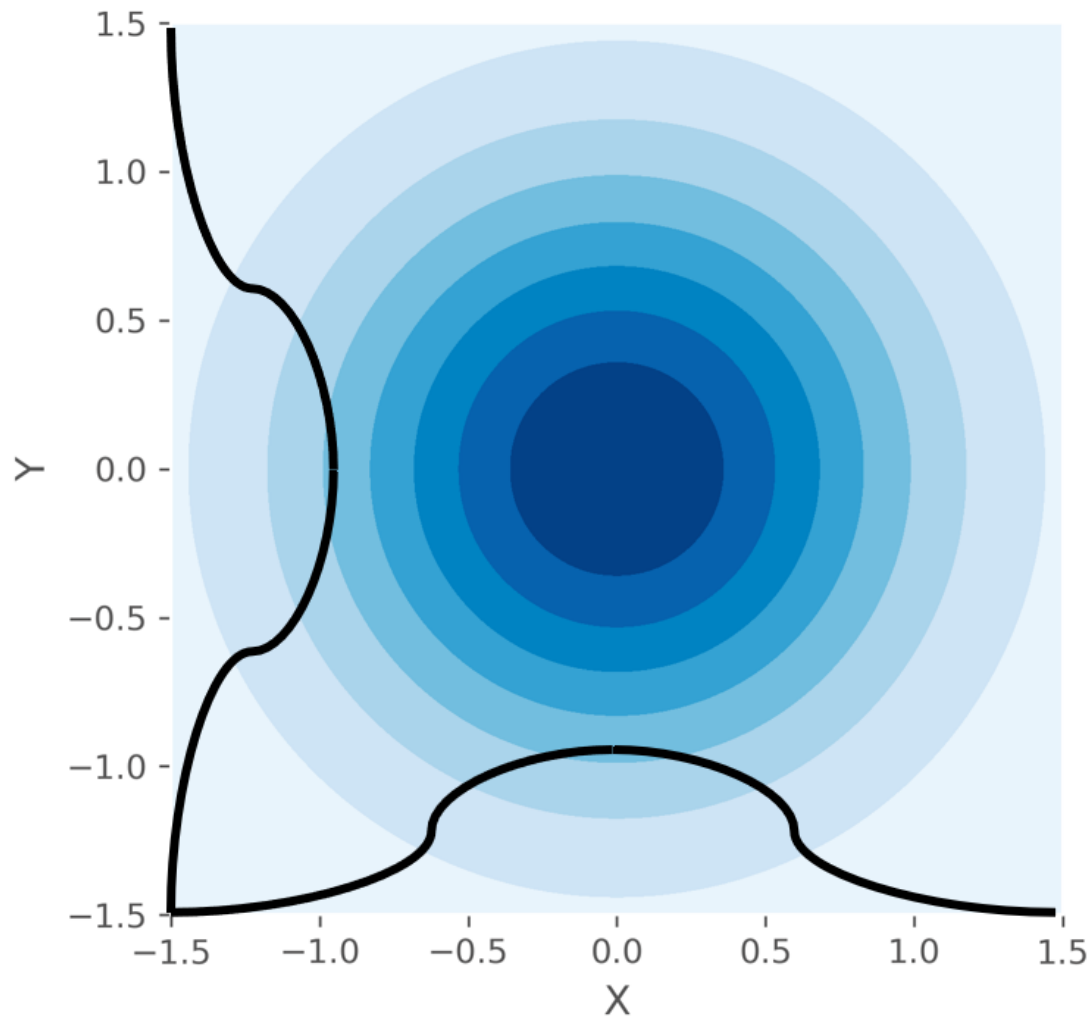


Gaussian probability distribution defined by mean & uncertainty



We can have multivariate Gaussian distributions

$$\mathcal{N}\left(\begin{bmatrix} 0.0 \\ 0.0 \end{bmatrix}, \begin{bmatrix} 0.5 & 0.0 \\ 0.0 & 0.5 \end{bmatrix}\right)$$



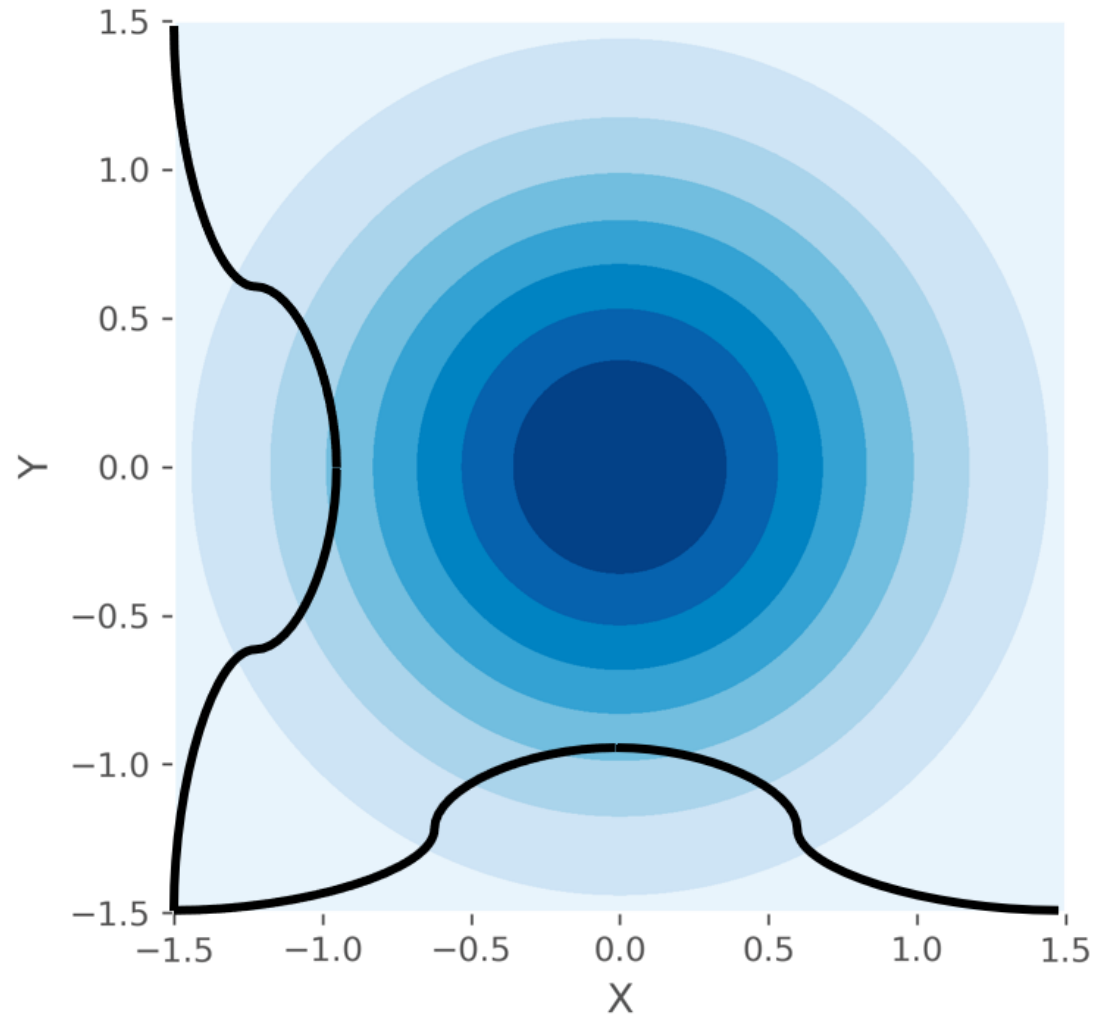
In Naïve Bayes, we assume all features are independent, so covariance matrix off-diagonal elements are all zero

$$\Sigma = \begin{pmatrix} \sigma_1^2 & 0 & \dots & 0 \\ 0 & \sigma_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_n^2 \end{pmatrix}$$

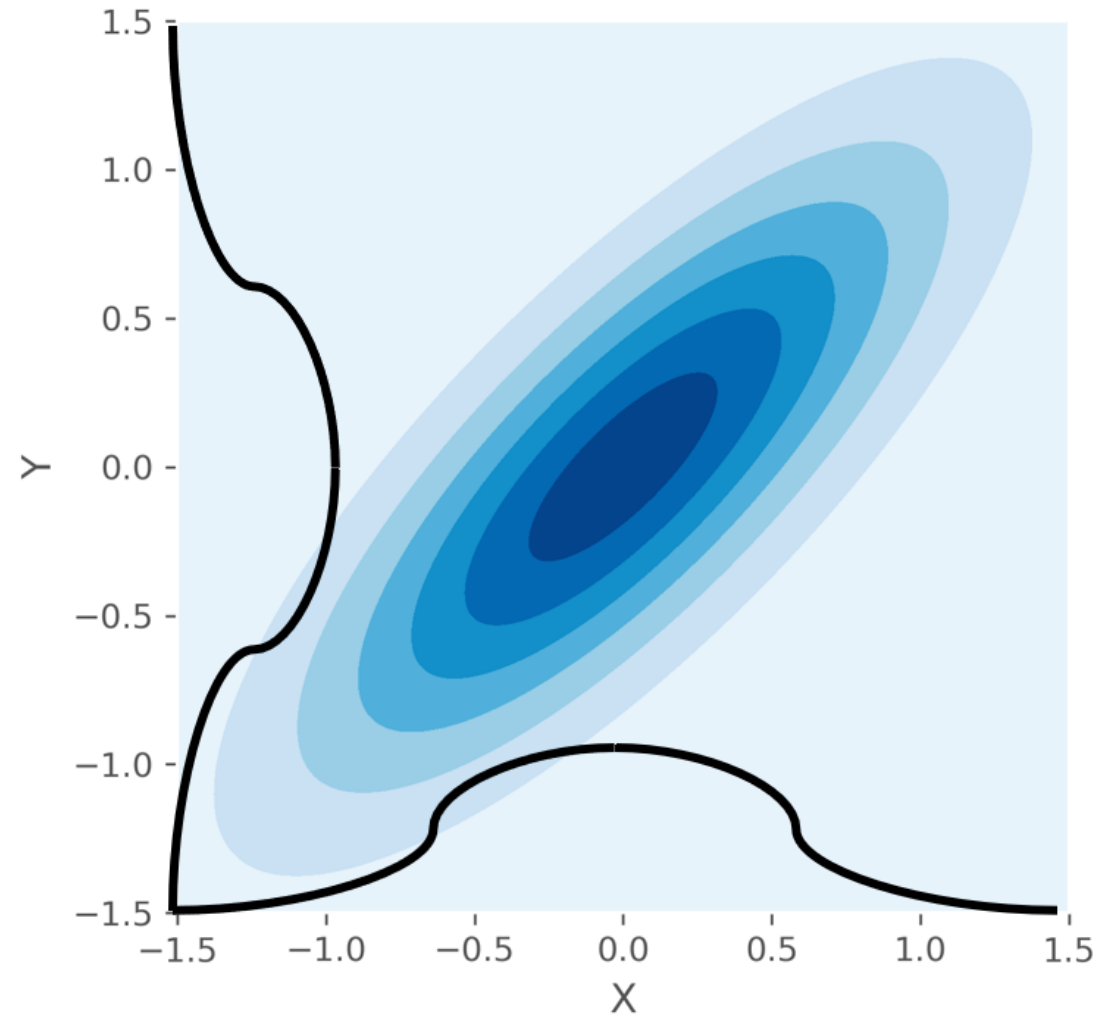


Our features could be correlated!

$$\mathcal{N}\left(\begin{bmatrix} 0.0 \\ 0.0 \end{bmatrix}, \begin{bmatrix} 0.5 & 0.0 \\ 0.0 & 0.5 \end{bmatrix}\right)$$



$$\mathcal{N}\left(\begin{bmatrix} 0.0 \\ 0.0 \end{bmatrix}, \begin{bmatrix} 0.5 & 0.4 \\ 0.4 & 0.5 \end{bmatrix}\right)$$



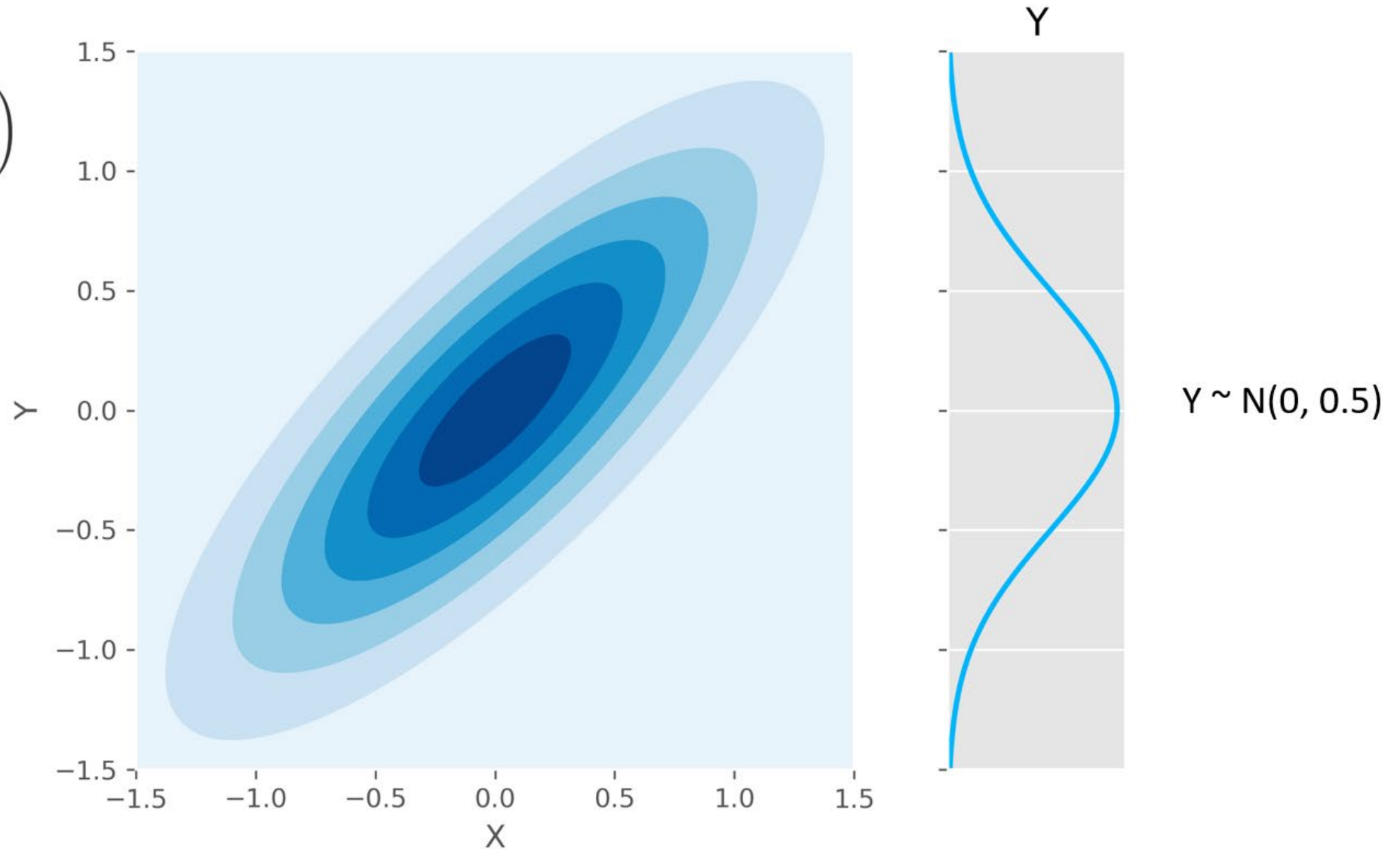
Gaussians can be **marginalized** and **conditioned**

$$p(y_1, y_2 | \mu, \Sigma) = N \left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \sigma_1^2 & \sigma_1 \sigma_2 \rho \\ \sigma_1 \sigma_2 \rho & \sigma_2^2 \end{bmatrix} \right)$$

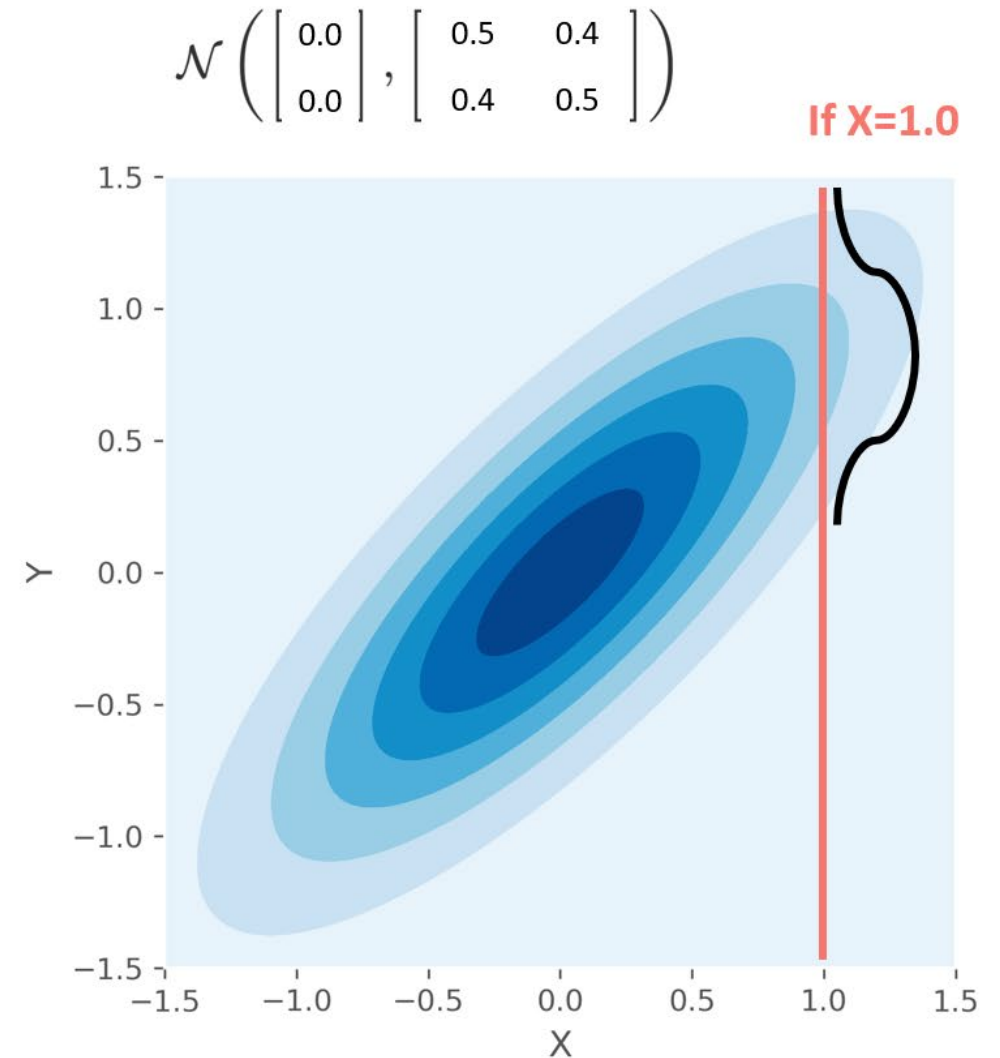
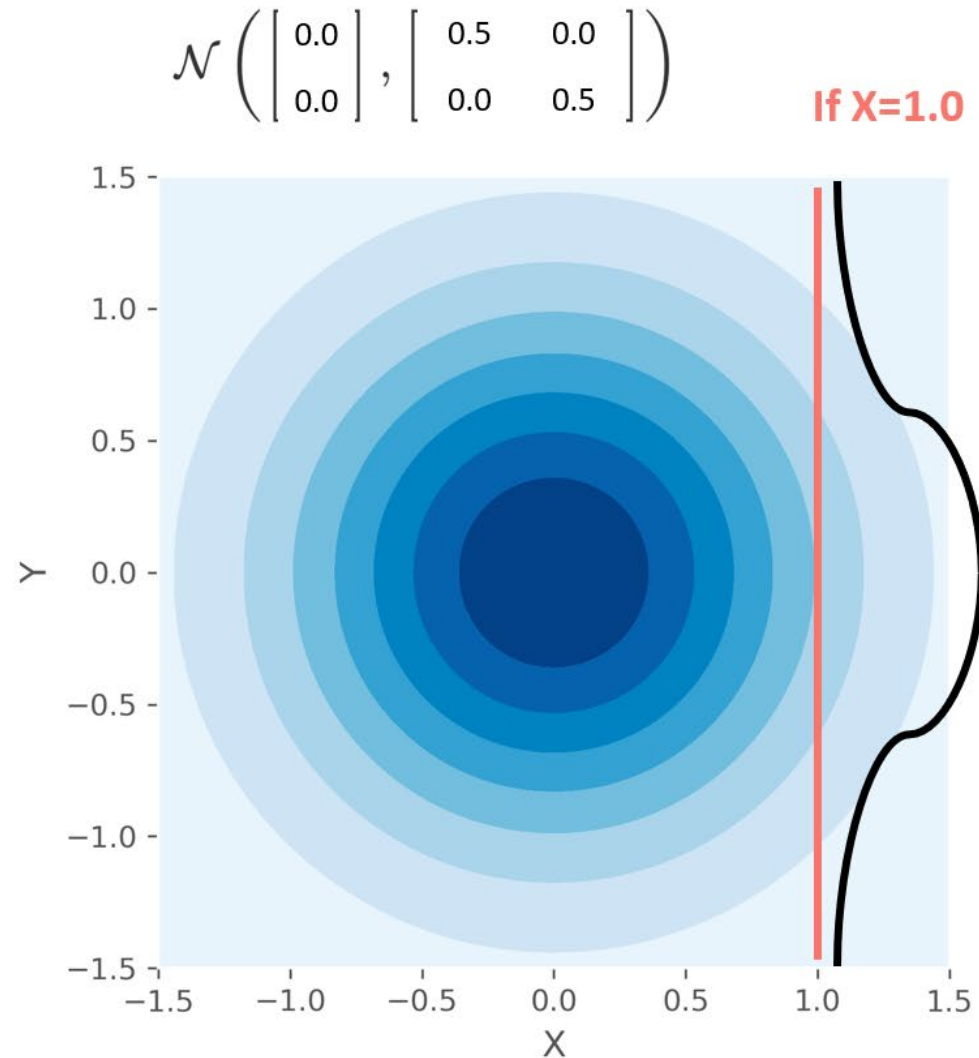
1. They are easy to **marginalize** (remove all parts of covariance matrix except what you want to keep)
 - if you want to ditch y_2 , then drop μ_2 , $\sigma_1 \sigma_2 \rho$, and σ_2^2
2. They are easy to **condition** (given the value of one value in our multivariate normal distribution, what do we expect some other unobserved value to be?)
 - $p(x|y) = N(\mu_x + \Sigma_{xx} \Sigma_y^{-1}(y - \mu_y), \Sigma_x - \Sigma_{xy} \Sigma_y^{-1} \Sigma_{xy}^T)$
Conditional mean, conditional covariance
 - Where μ_x is marginal mean (easy) and Σ_x is marginal covariance each minus some easy to calculate terms

Marginalization allows us to extract individual distributions

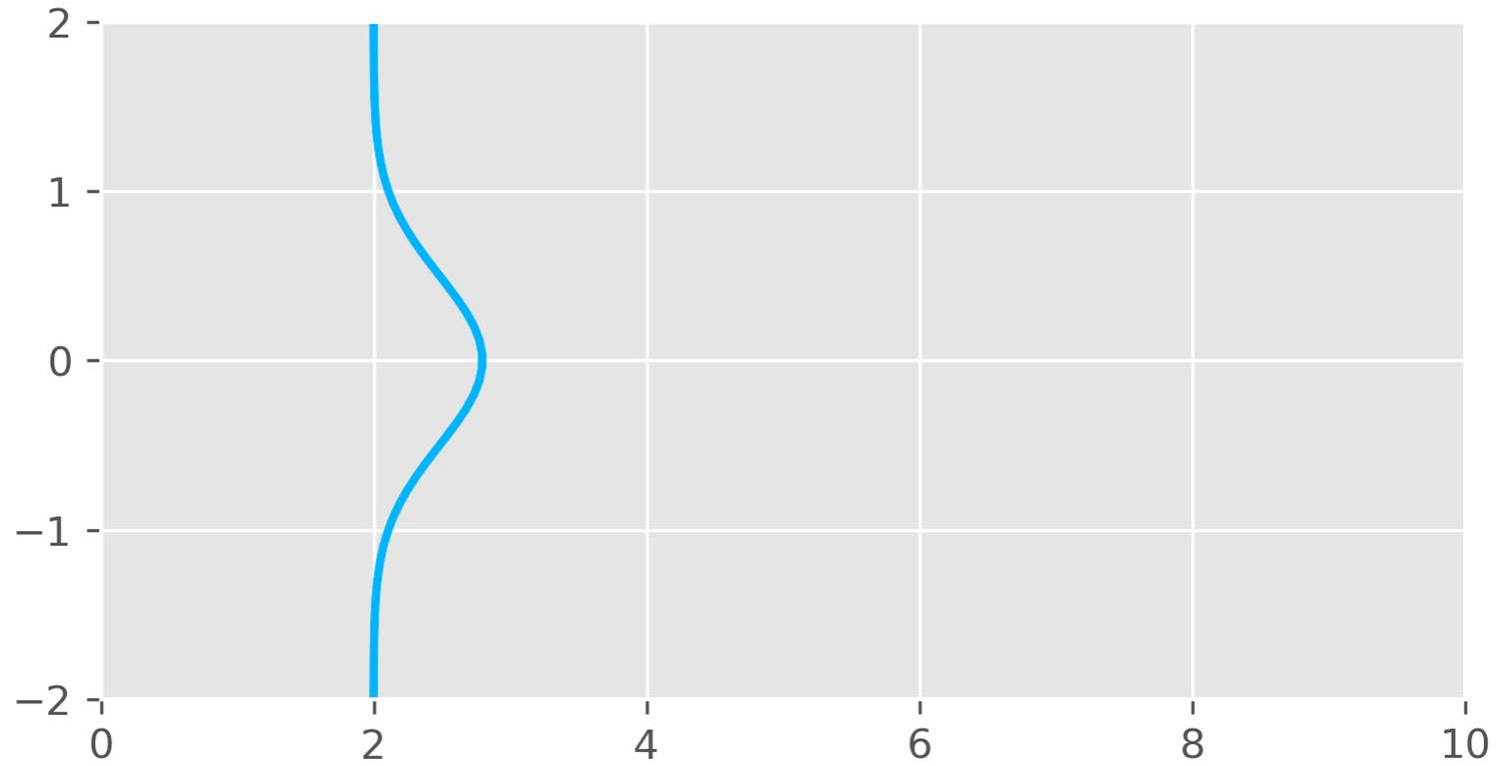
$$\mathcal{N}\left(\begin{bmatrix} 0.0 \\ 0.0 \end{bmatrix}, \begin{bmatrix} 0.5 & 0.4 \\ 0.4 & 0.5 \end{bmatrix}\right)$$



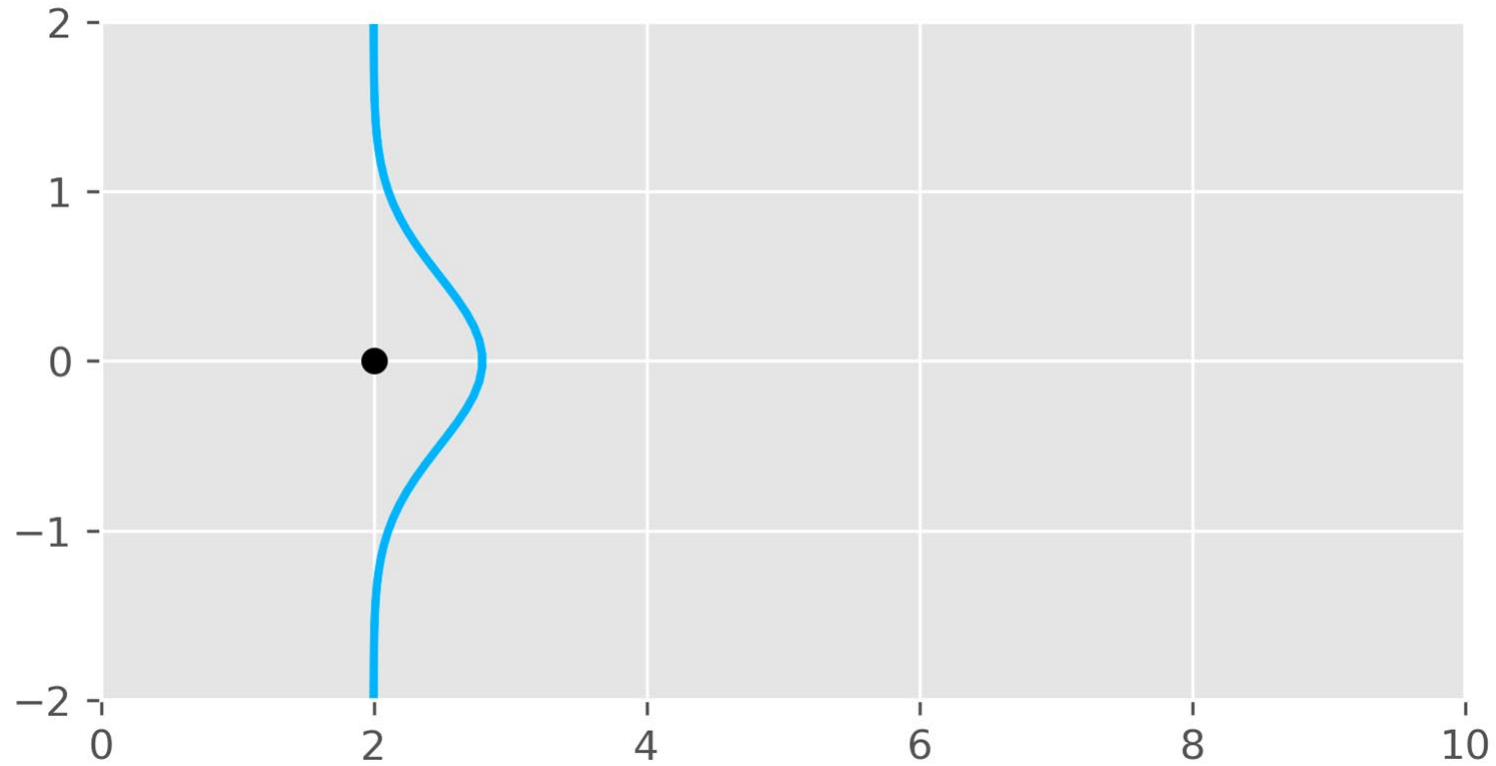
Conditioning tells us how variables correlate to each other



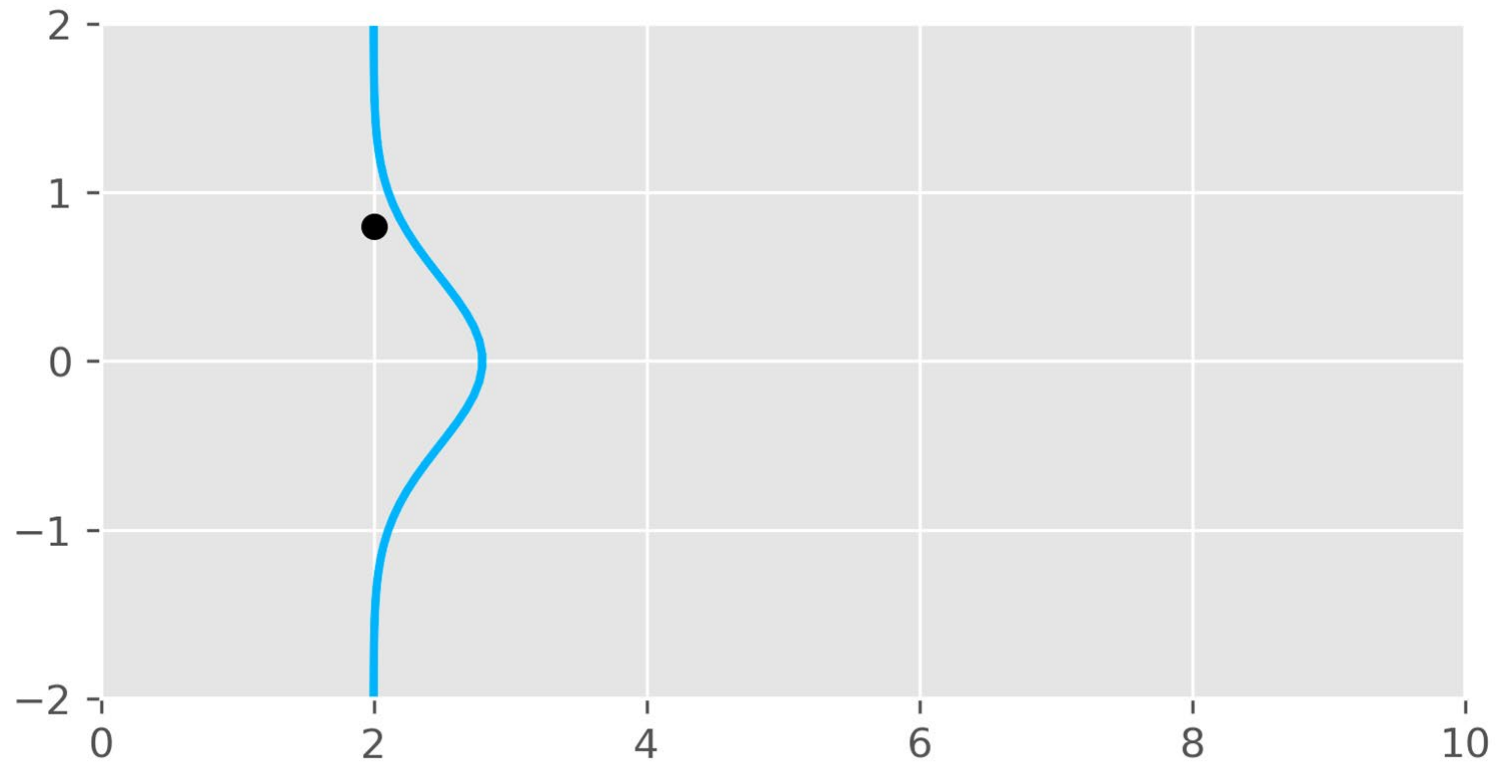
Gaussian process is a collection of infinitely many gaussians



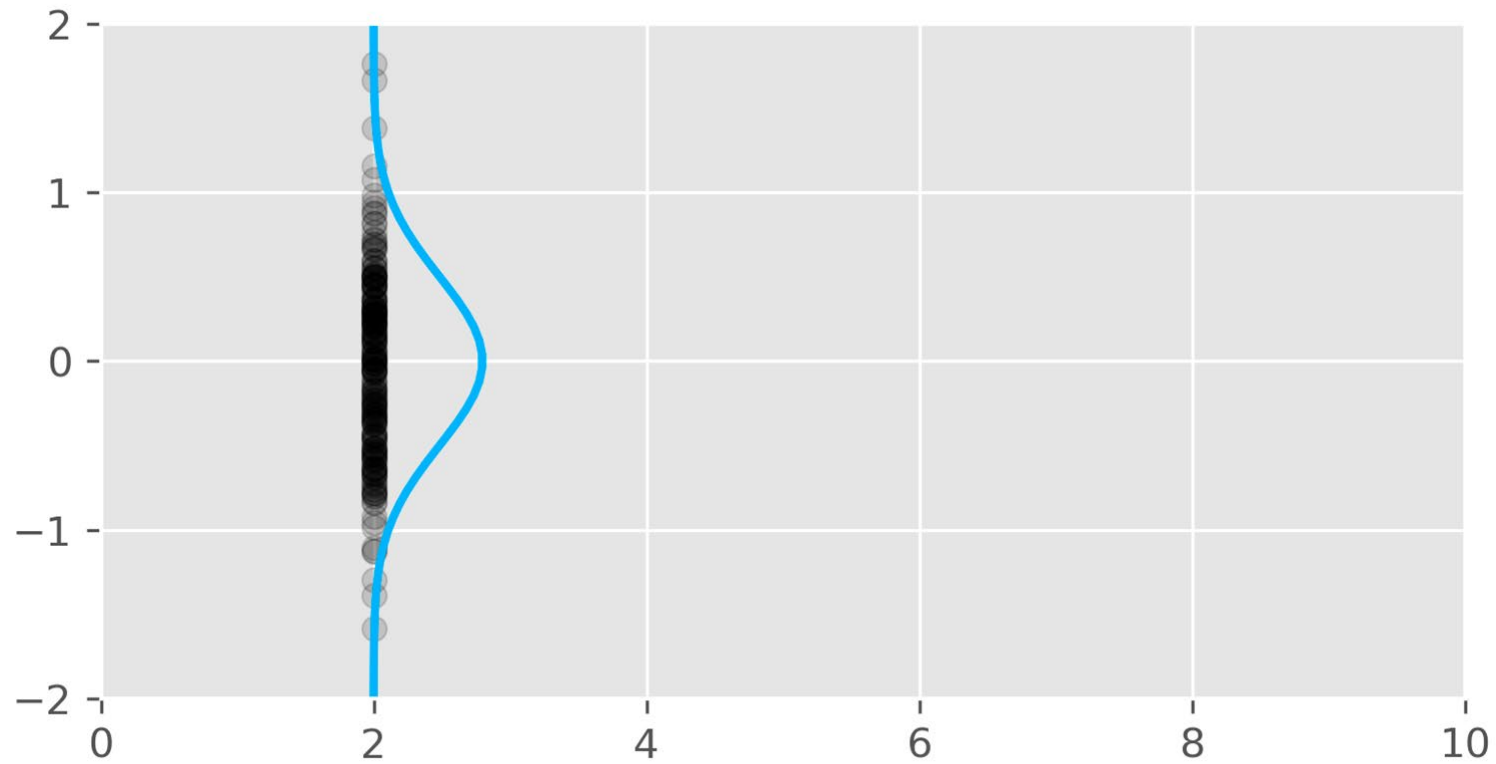
Gaussian process is a collection of infinitely many gaussians



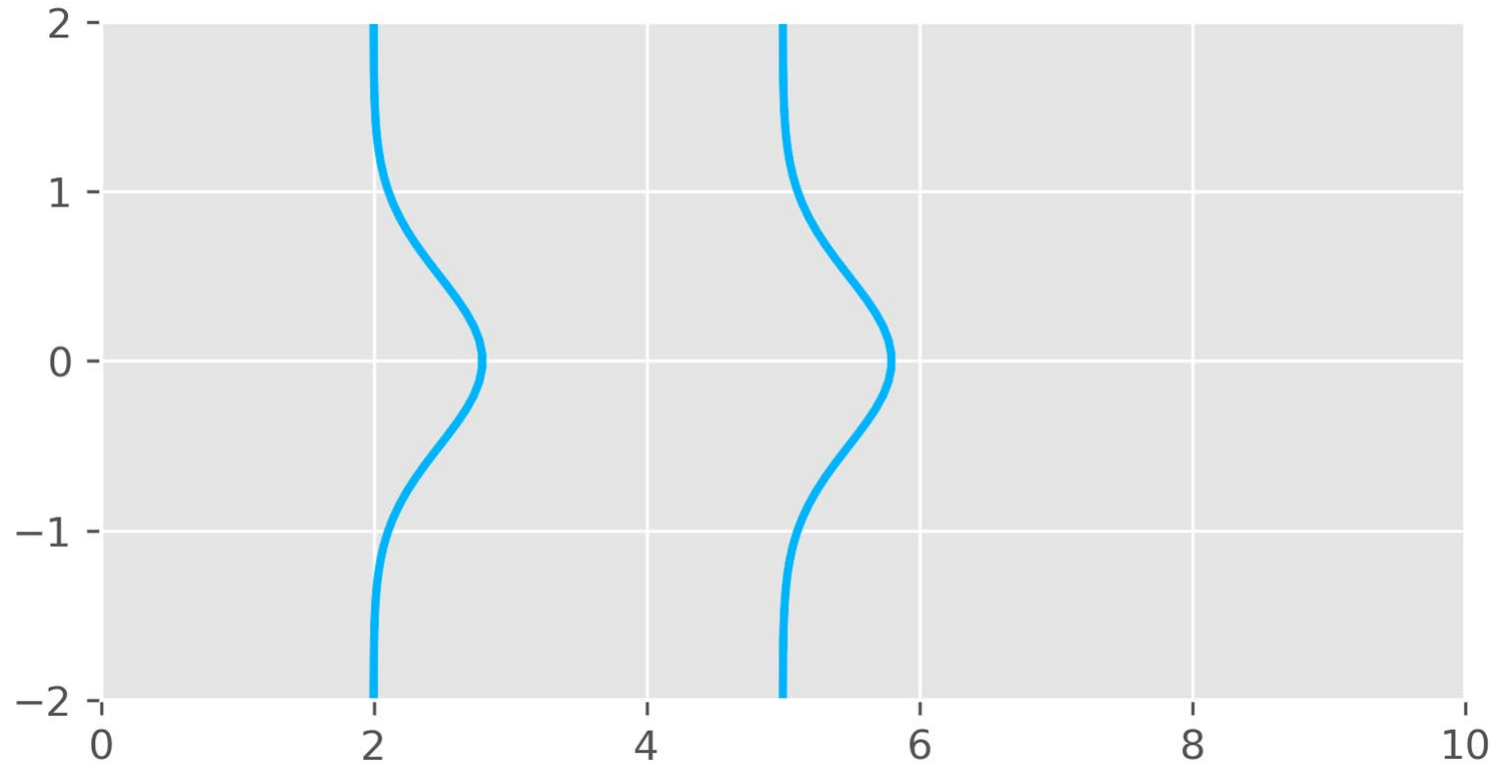
Gaussian process is a collection of infinitely many gaussians



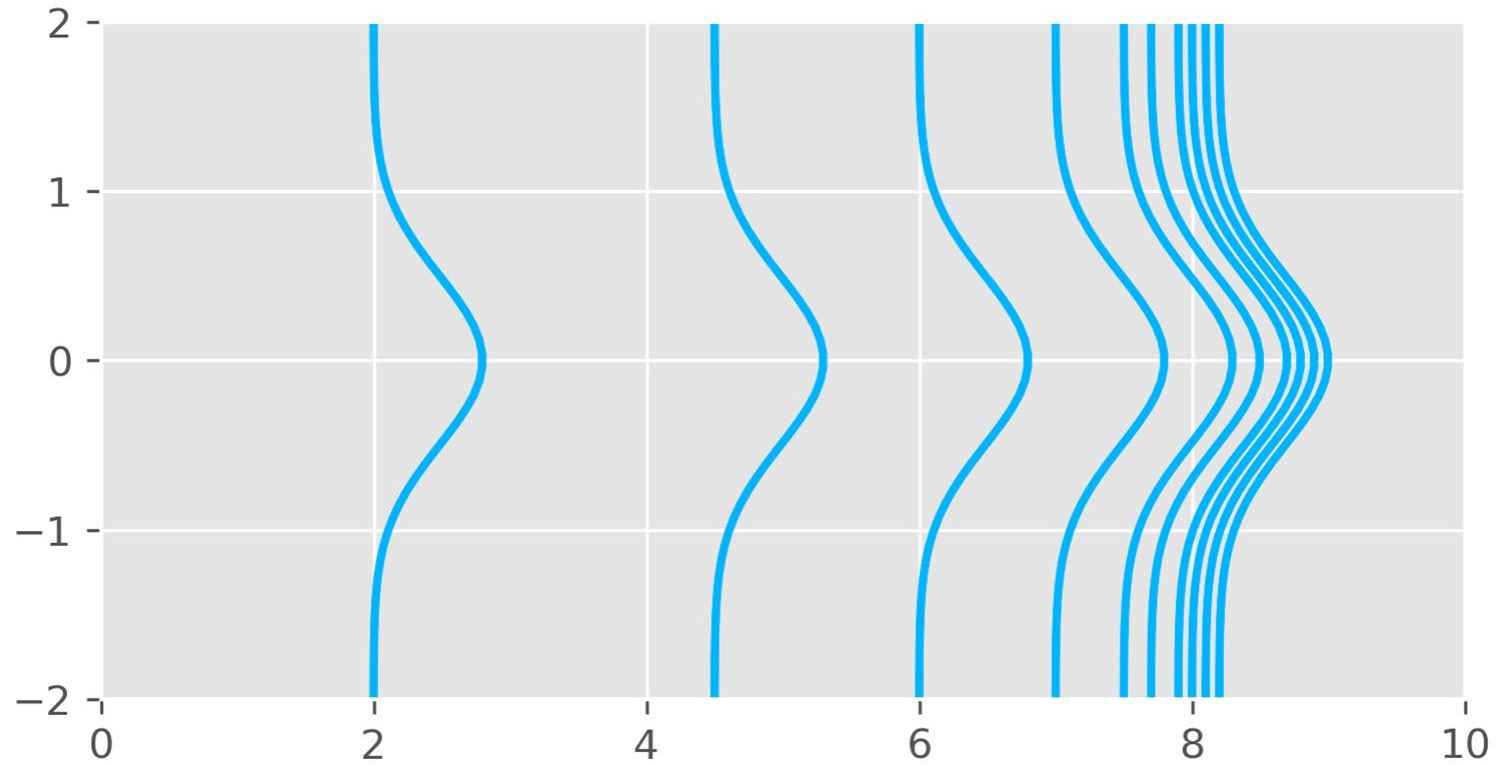
Gaussian process is a collection of infinitely many gaussians



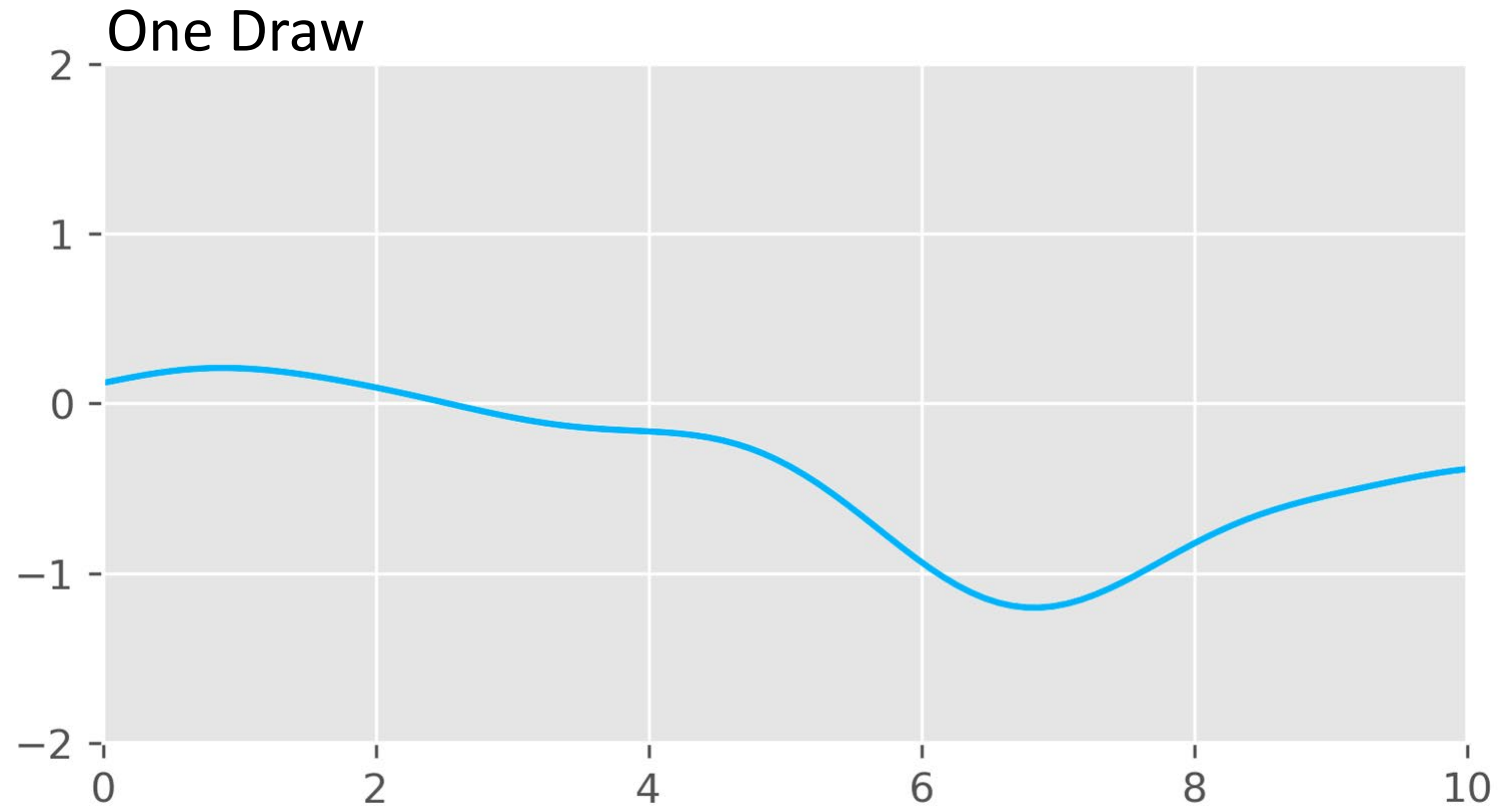
Gaussian process is a collection of infinitely many gaussians



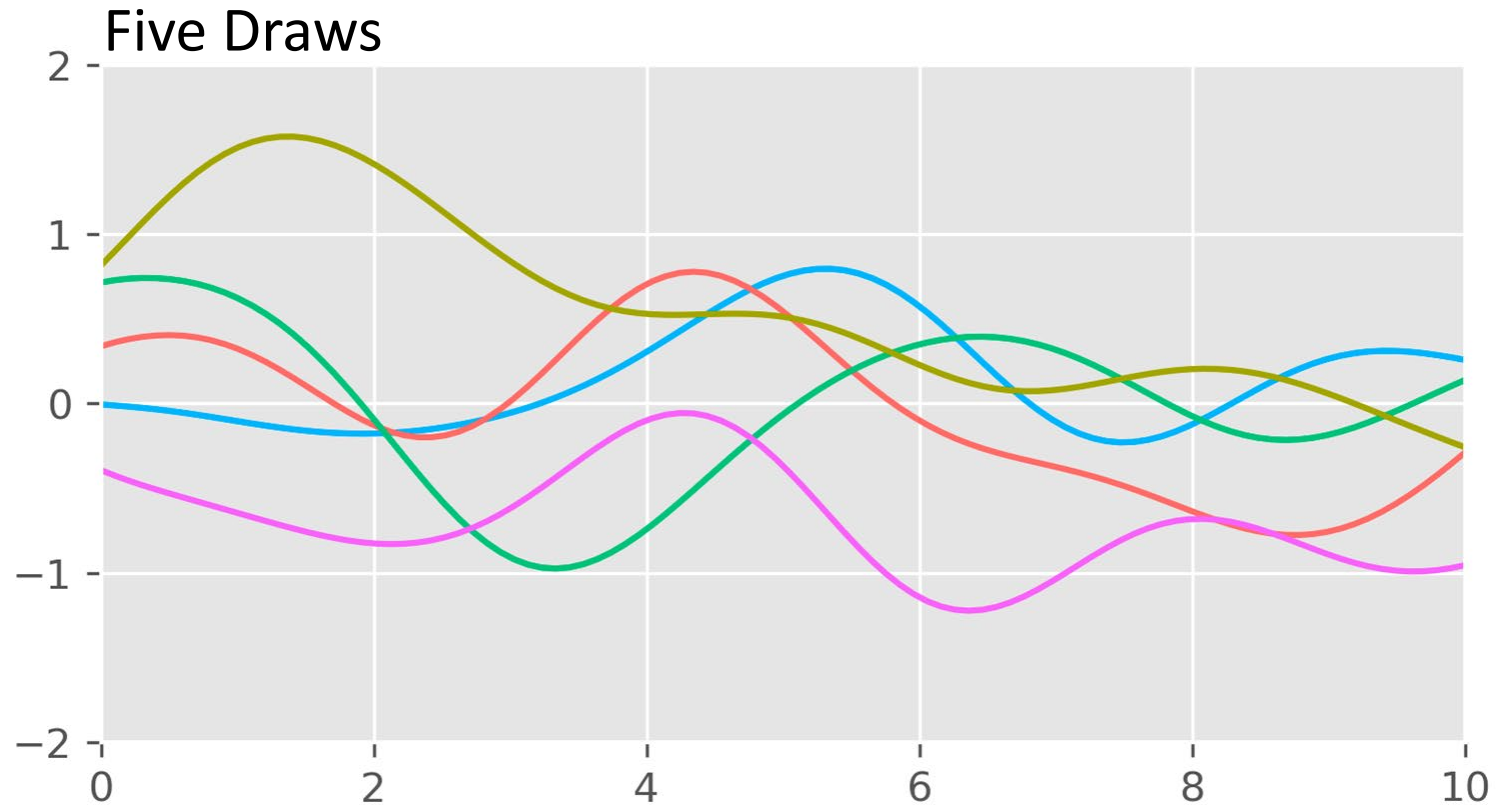
Gaussian process is a collection of infinitely many gaussians



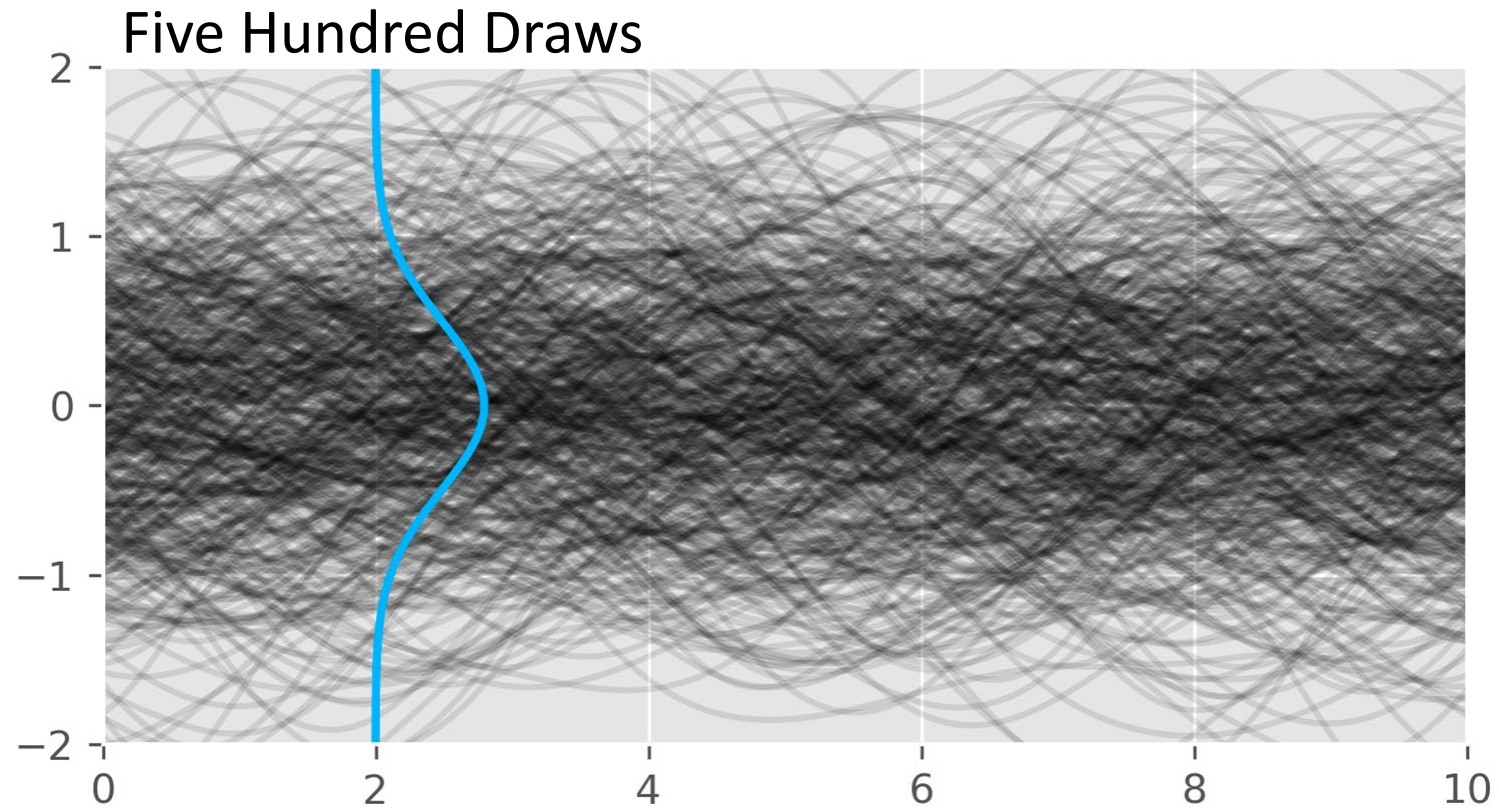
Drawing from marginalized gaussians produces a single function



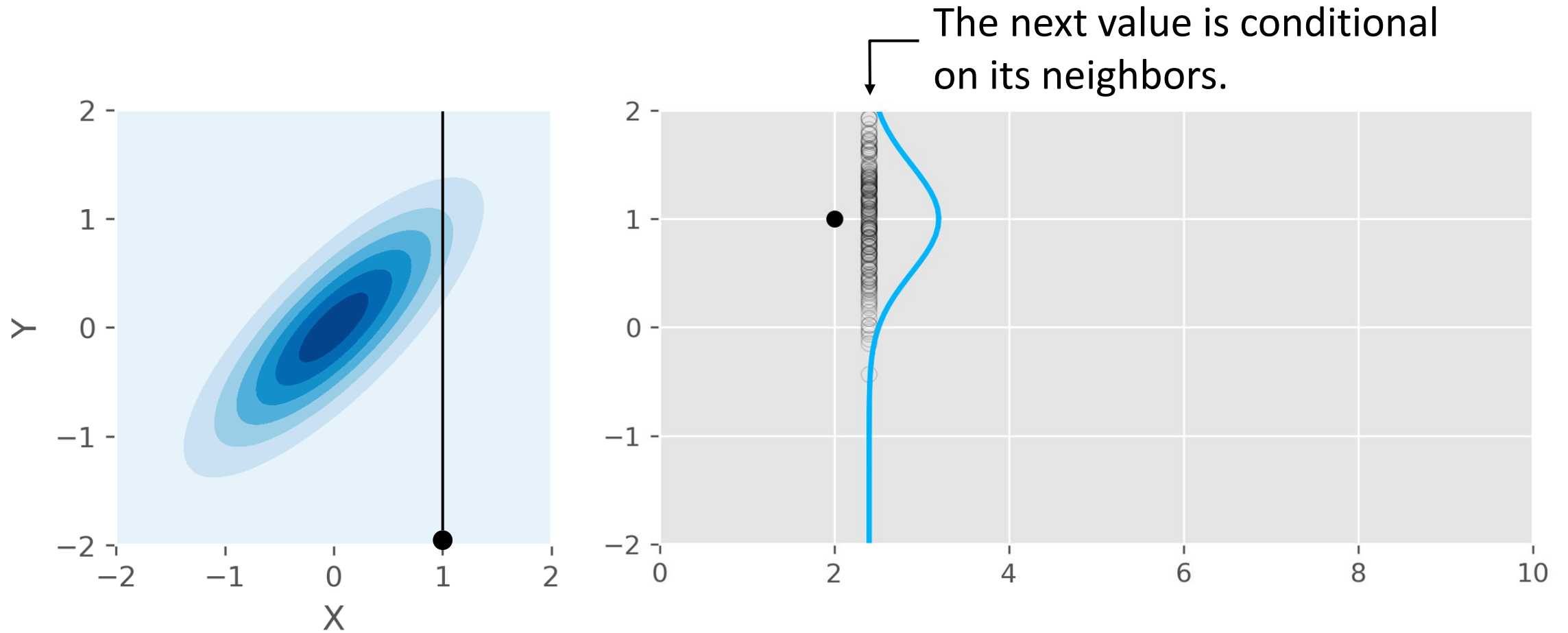
Drawing from marginalized gaussians can produce many functions



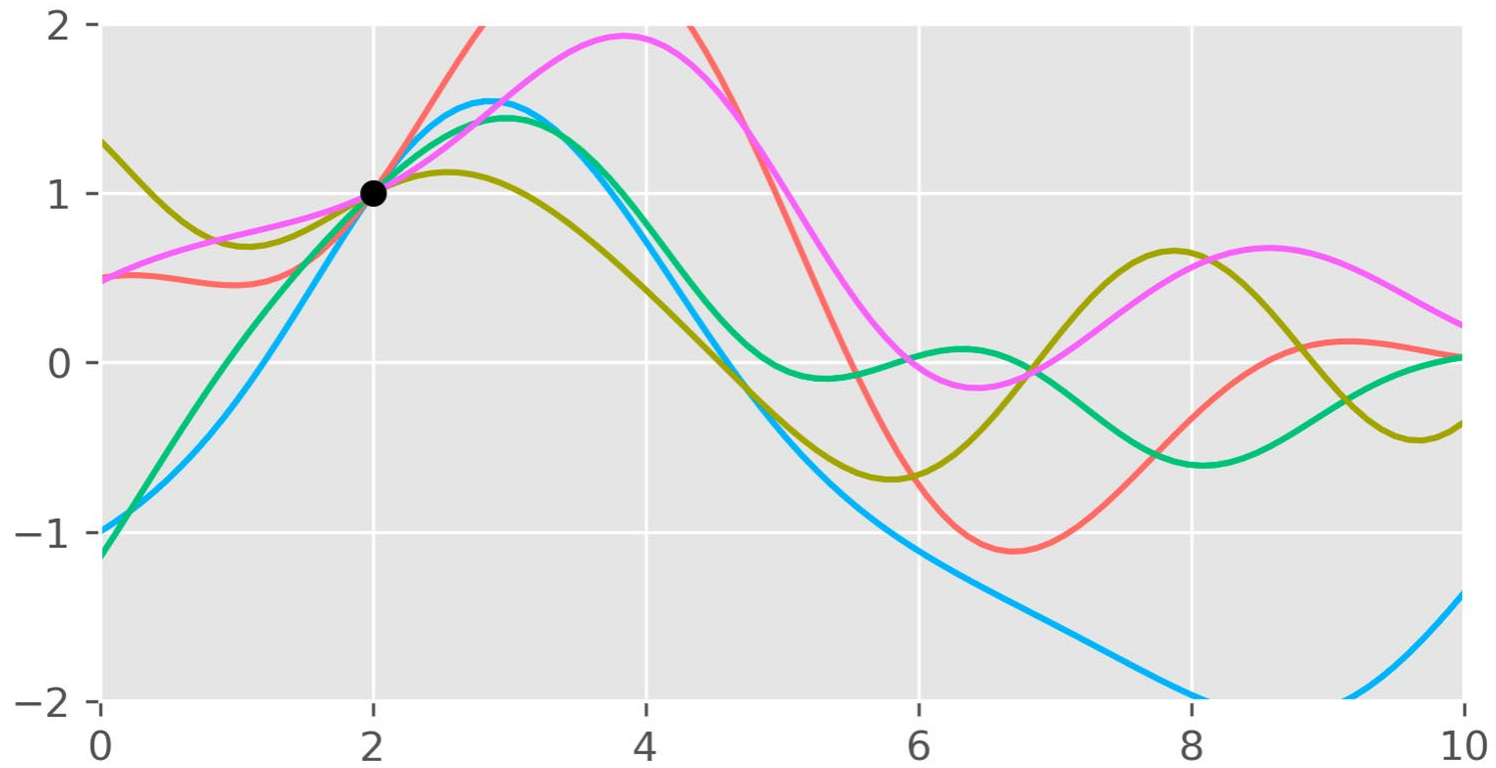
Drawing from marginalized gaussians produces infinite functions



Conditioning restricts neighboring points to a subset of values

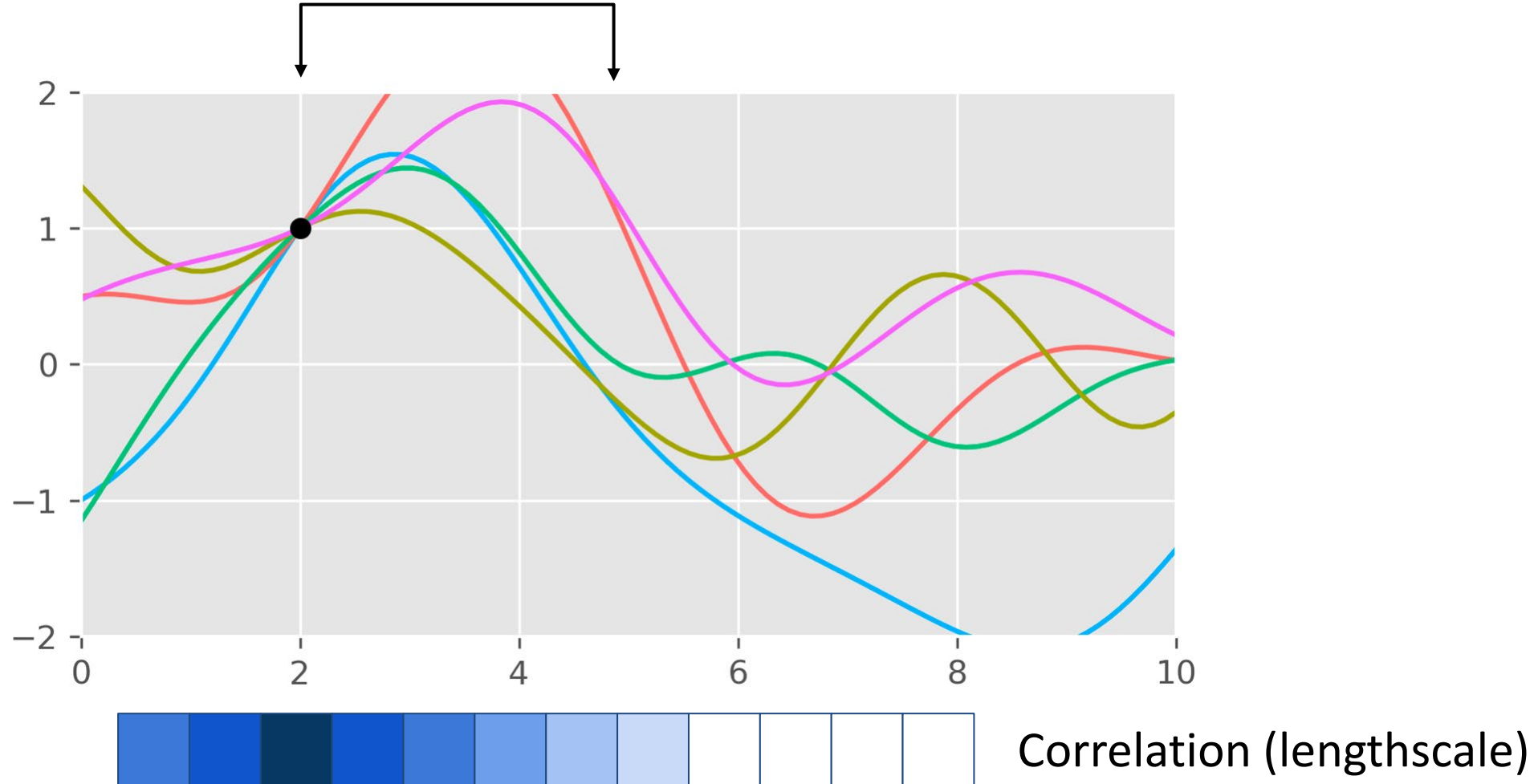


Function draws can then be conditioned on observed data!

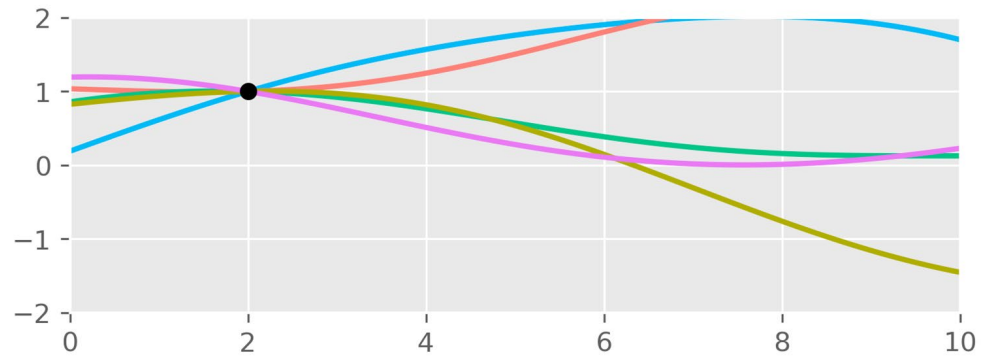


The influence of a point is governed by the length scale parameter

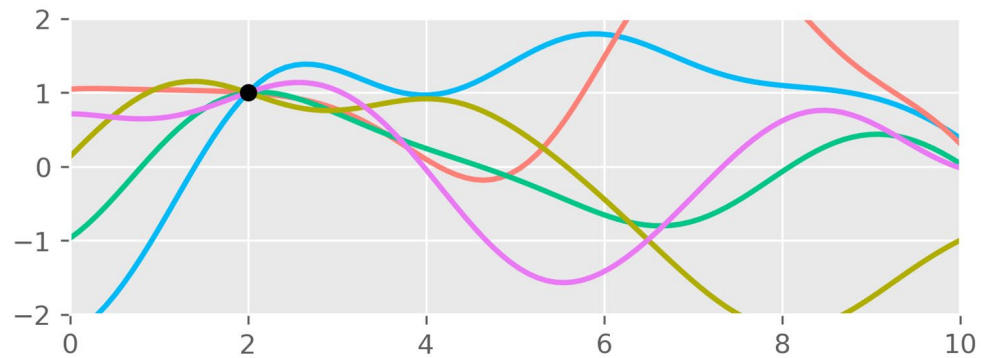
How much should this observation influence this point?



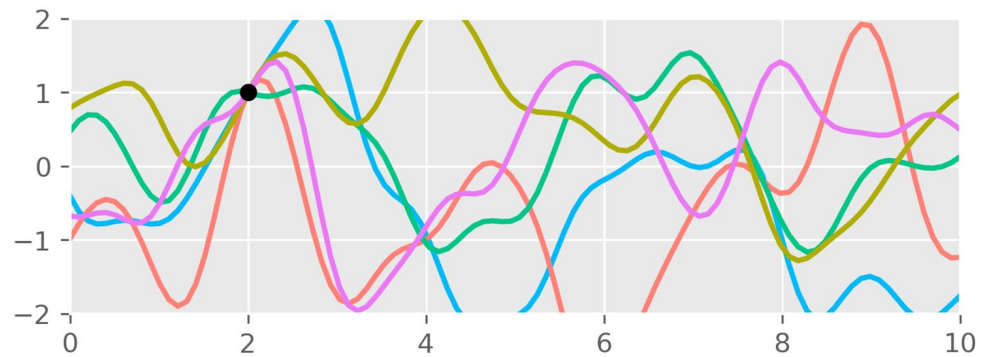
Longer length scales give more influence



$L = 5$

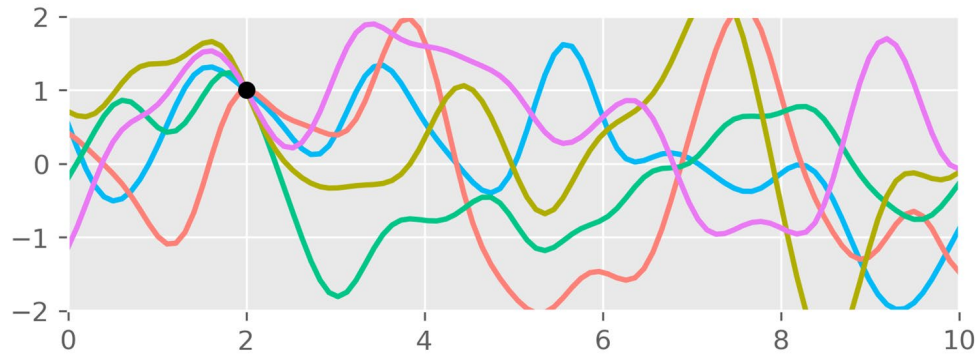


$L = 1.5$

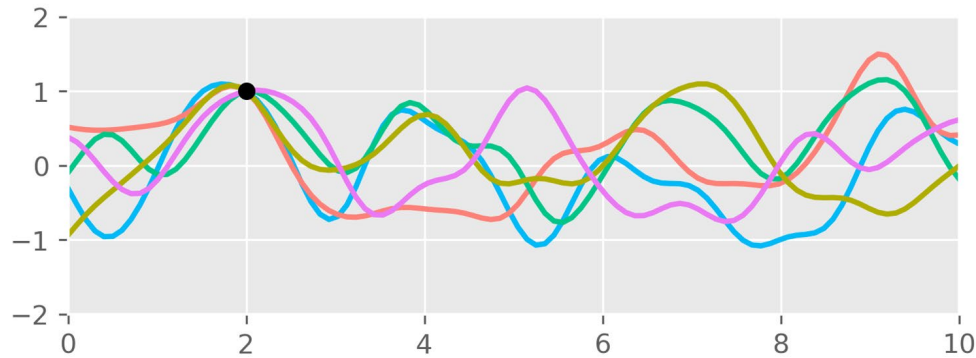


$L = 0.5$

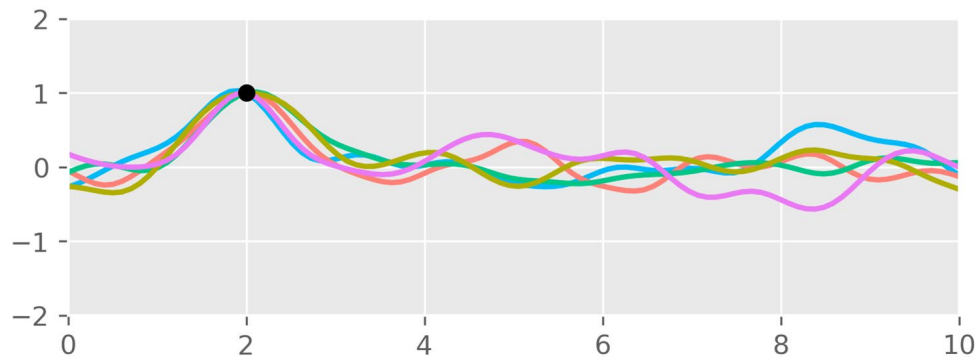
The variance parameter controls the spread of possible functions



$\sigma = 1.0$



$\sigma = 0.25$



$\sigma = 0.05$

A proper GP is defined by a mean and covariance function

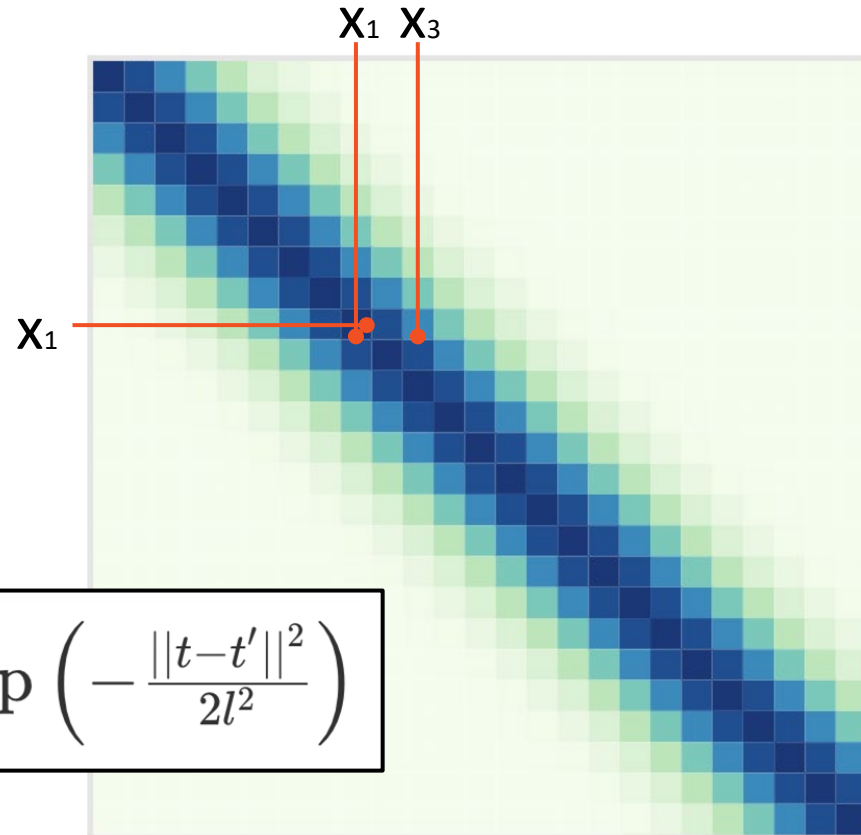
$$f(x) \sim \text{GP}(m, k)$$



Usually a constant, but
it doesn't have to be!



Defines how two points in space are
correlated.



RBK Kernel

$$k(t, t') = \sigma^2 \exp \left(-\frac{\|t - t'\|^2}{2l^2} \right)$$

Figure referenced from:

<https://distill.pub/2019/visual-exploration-gaussian-processes/>

There are tons of different covariance functions

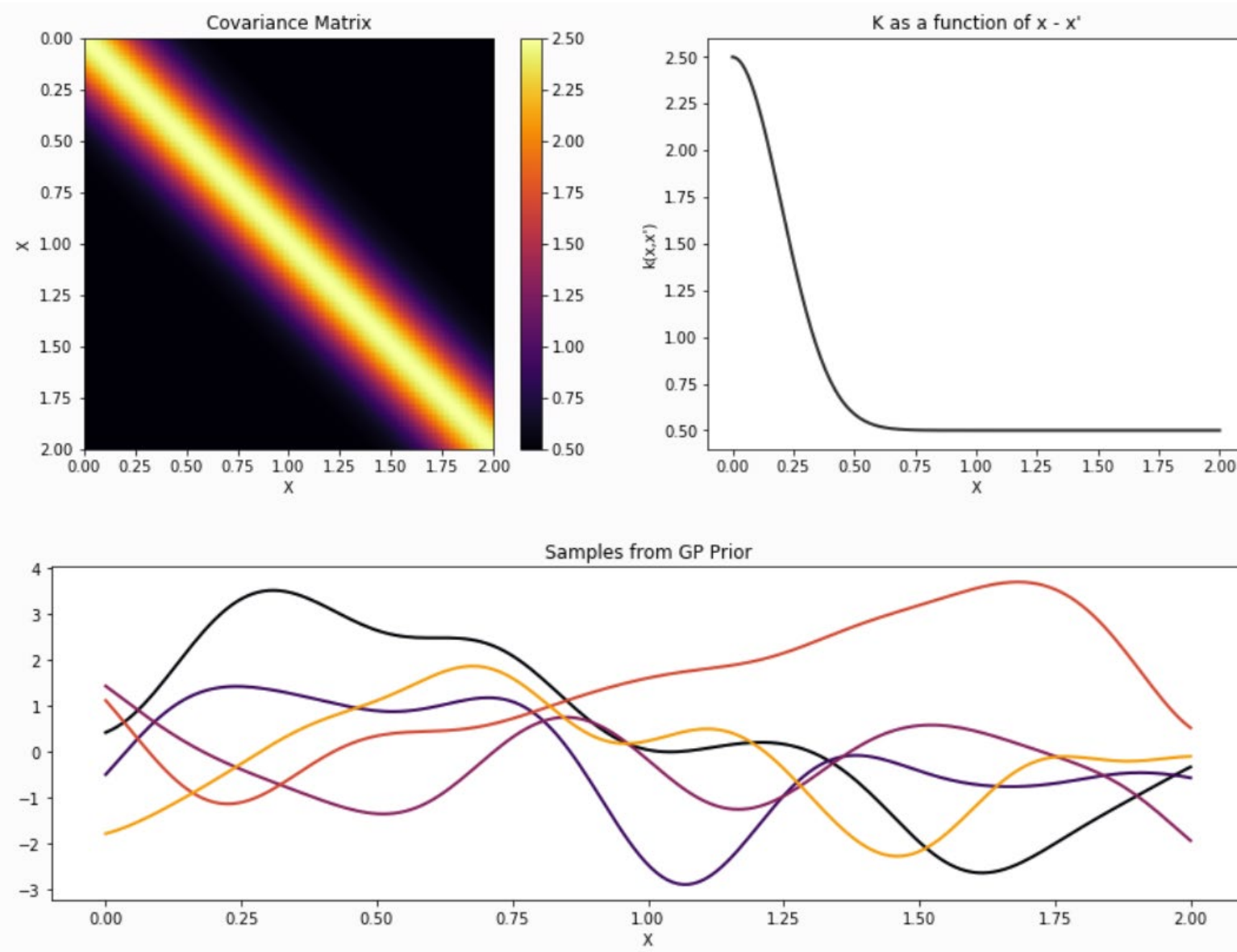


Figure Referenced From
https://www.pymc.io/projects/examples/en/latest/gaussian_processes/GP-MeansAndCovs.html

There are tons of different covariance functions

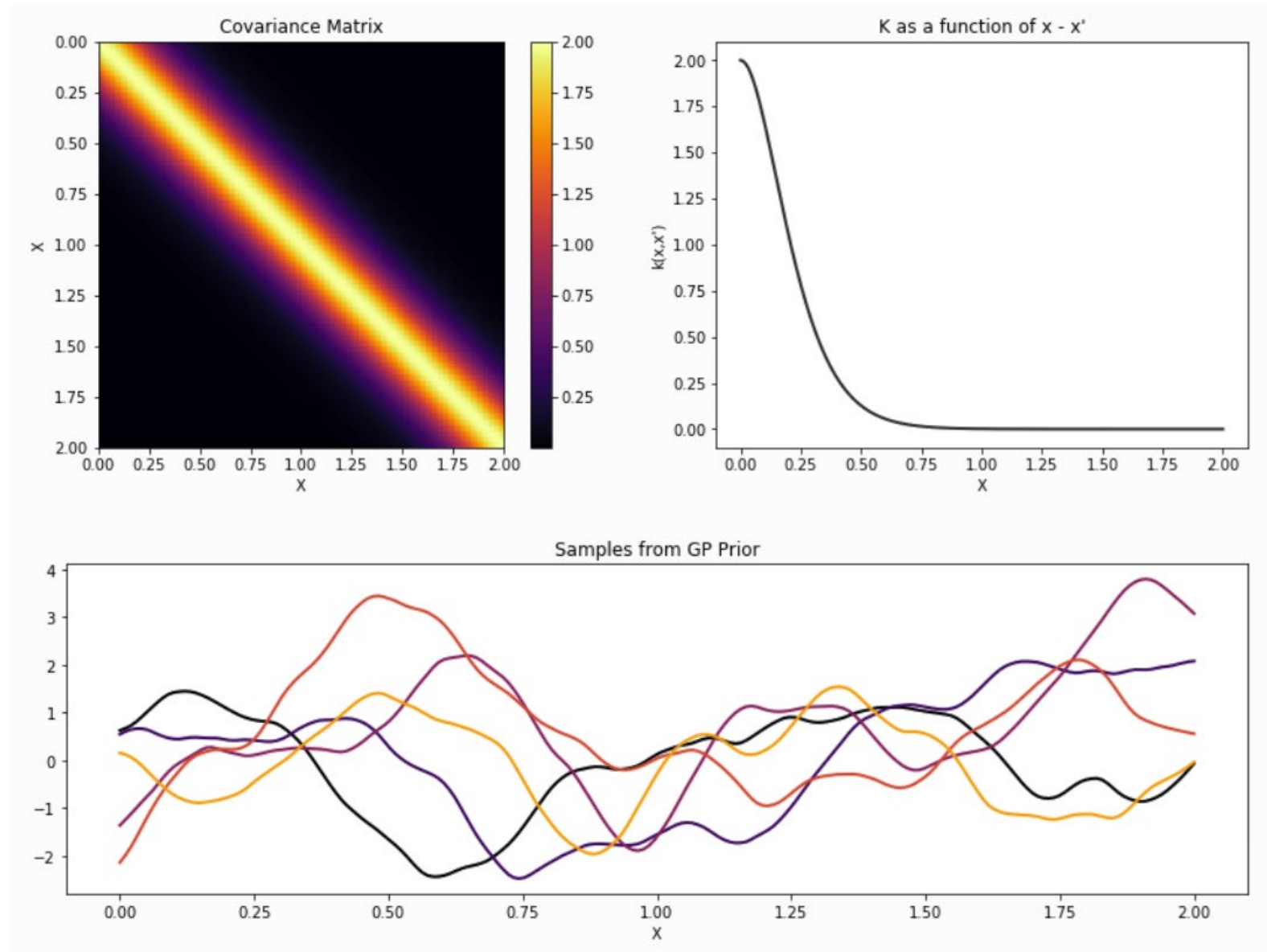


Figure Referenced From
https://www.pymc.io/projects/examples/en/latest/gaussian_processes/GP-MeansAndCovs.html

There are tons of different covariance functions

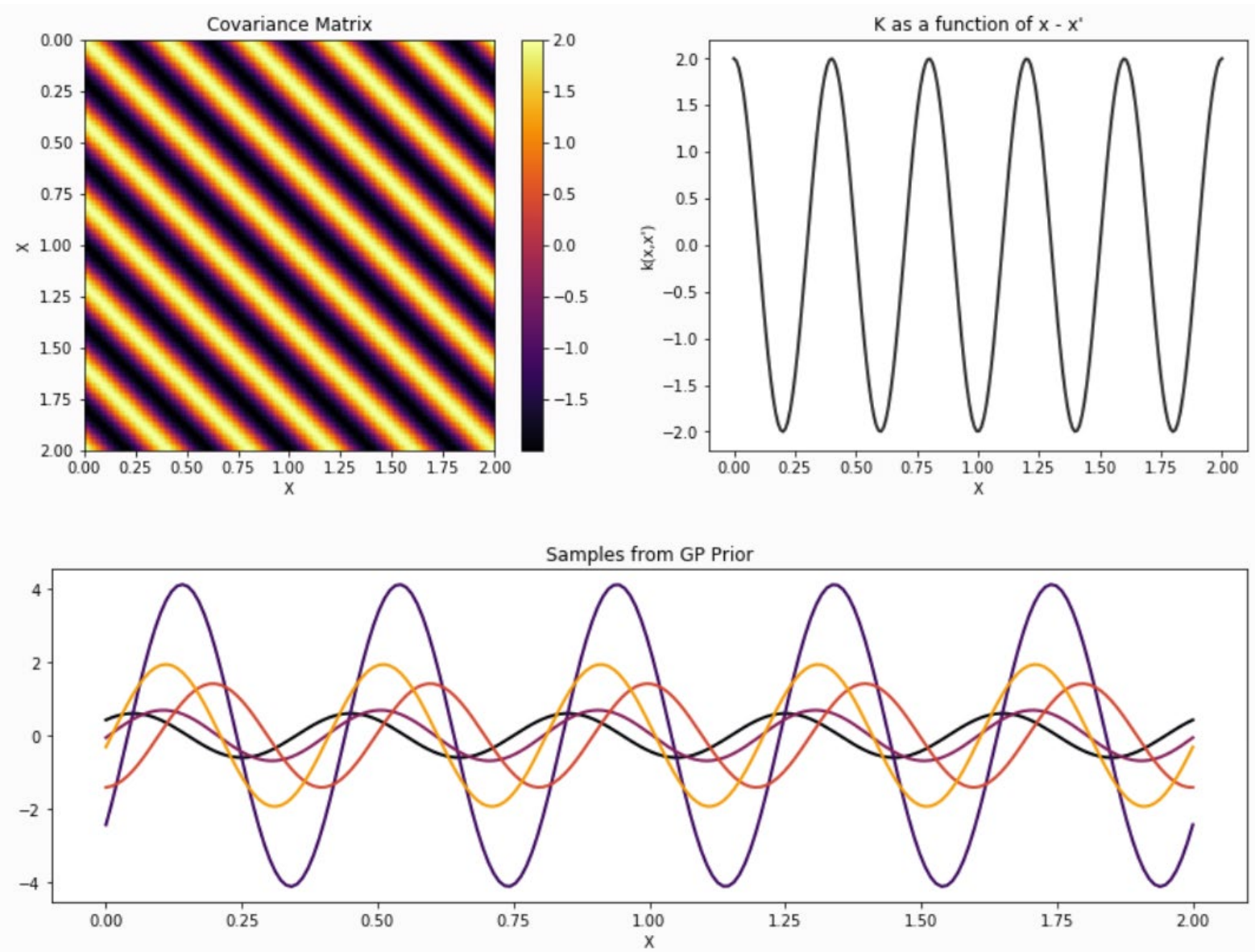


Figure Referenced From
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There are tons of different covariance functions

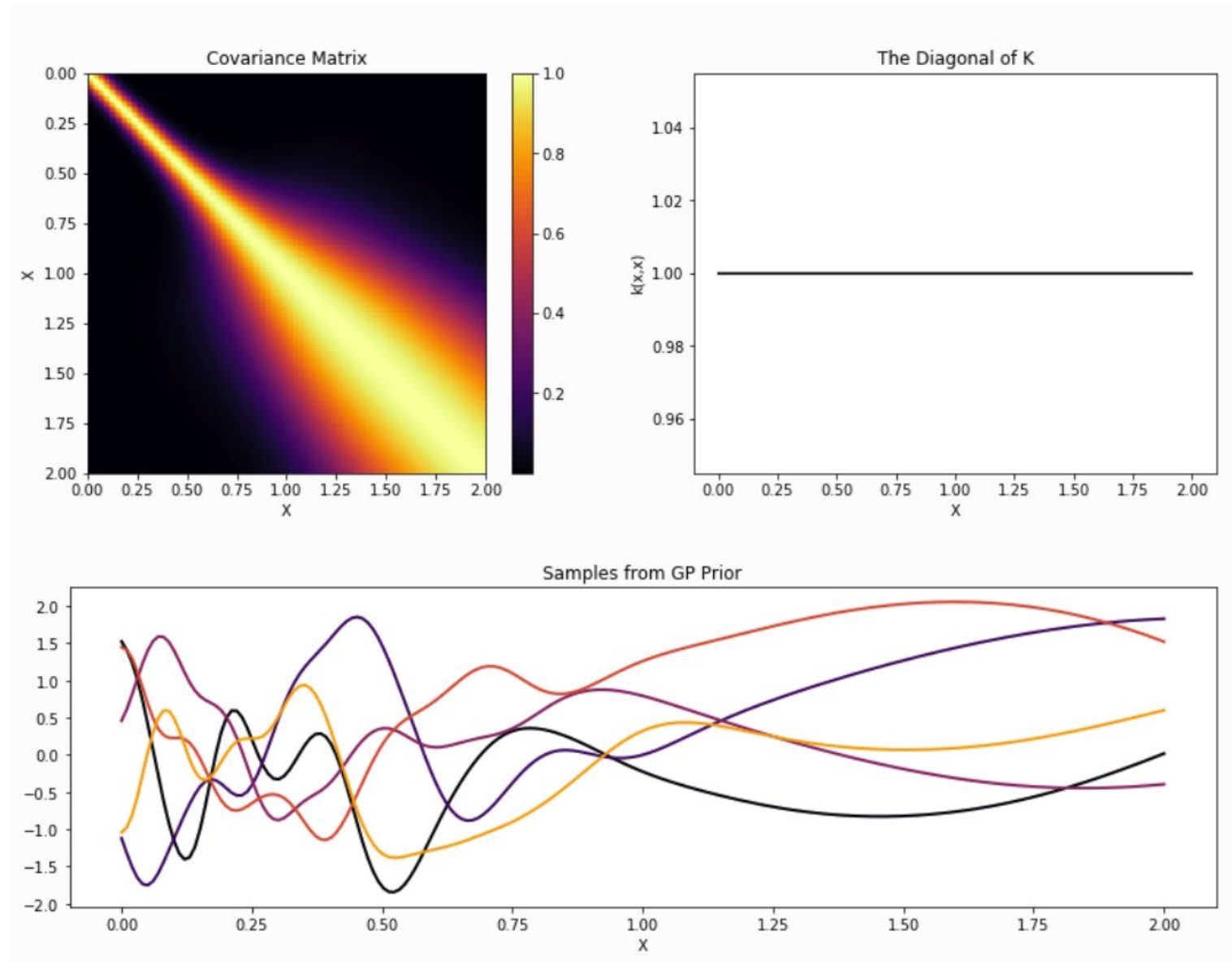
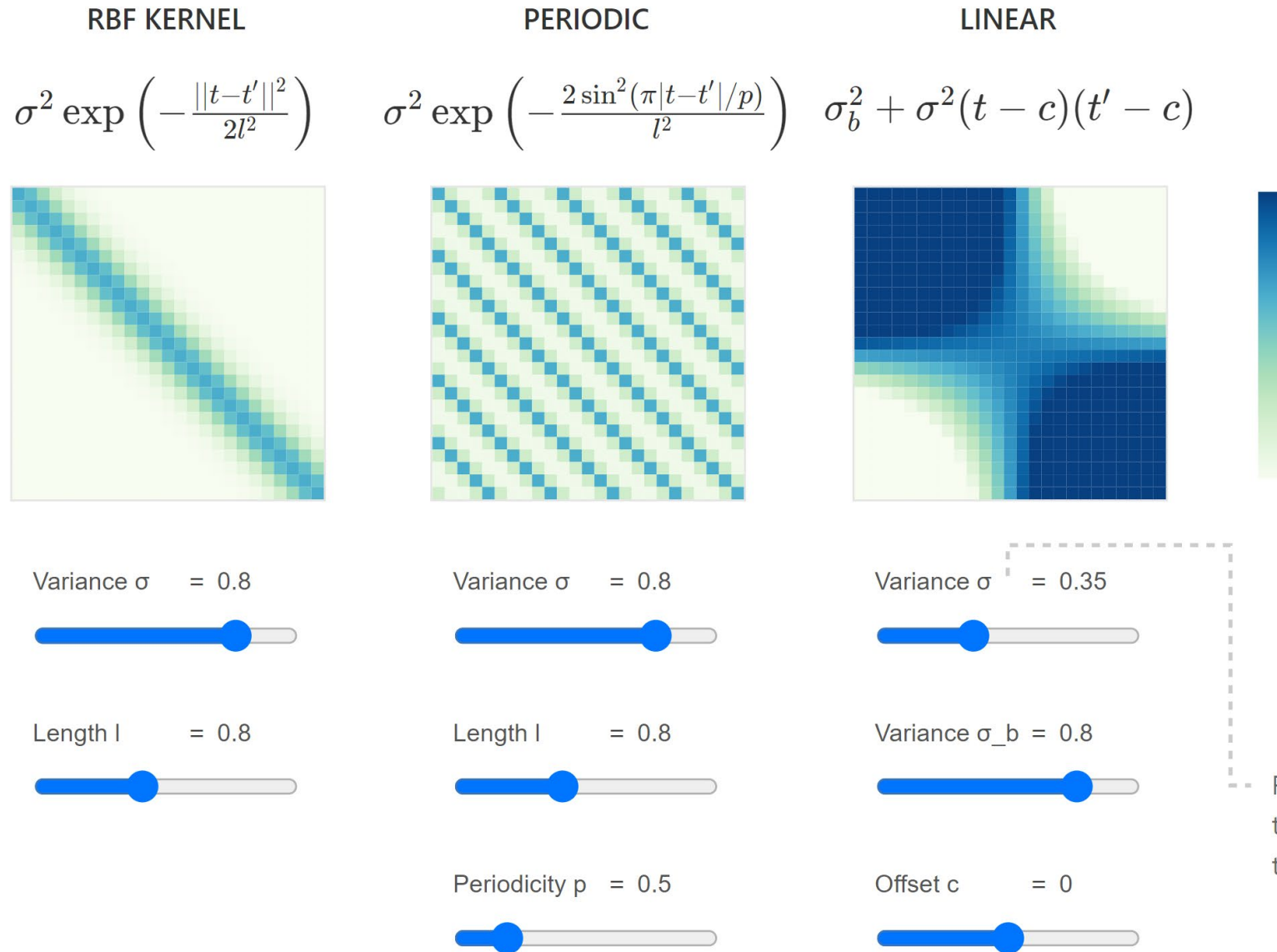


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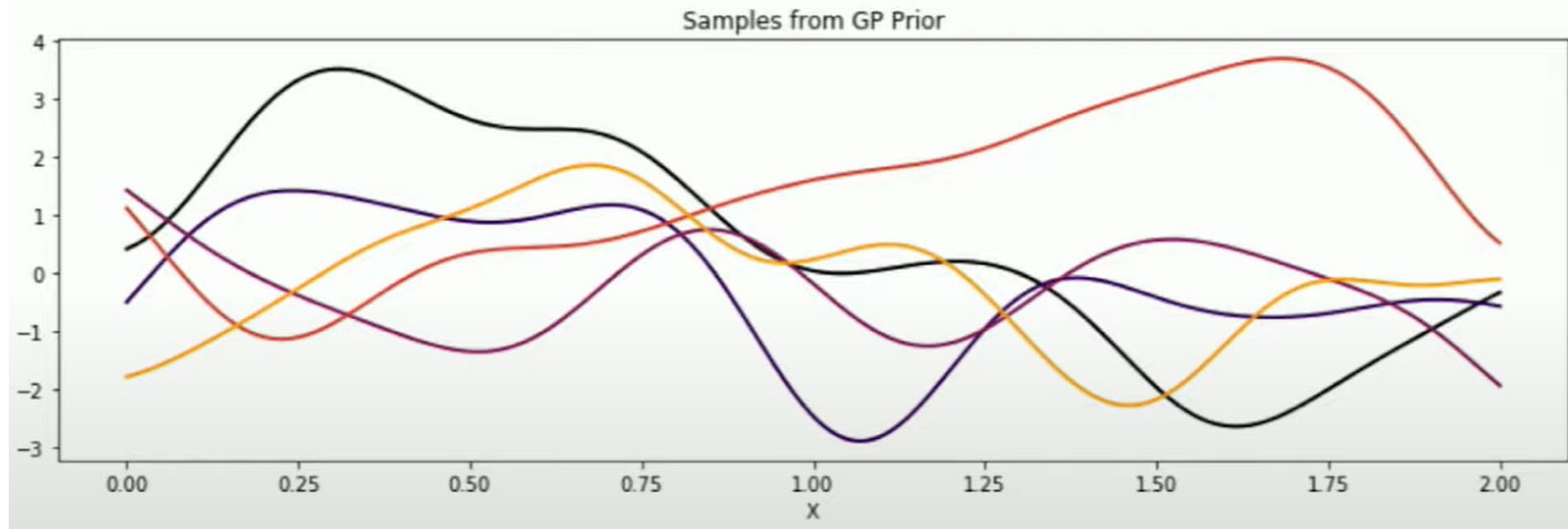
It's tricky to know which function to use to model covariance 🤔



For the **Linear** kernel the parameter **Variance σ** determines the average distance away from the function's mean.

Consider the many covariance functions just from exponential quadratic

$$k(x, x') = \exp \left[-\frac{(x - x')^2}{2\ell^2} \right]$$

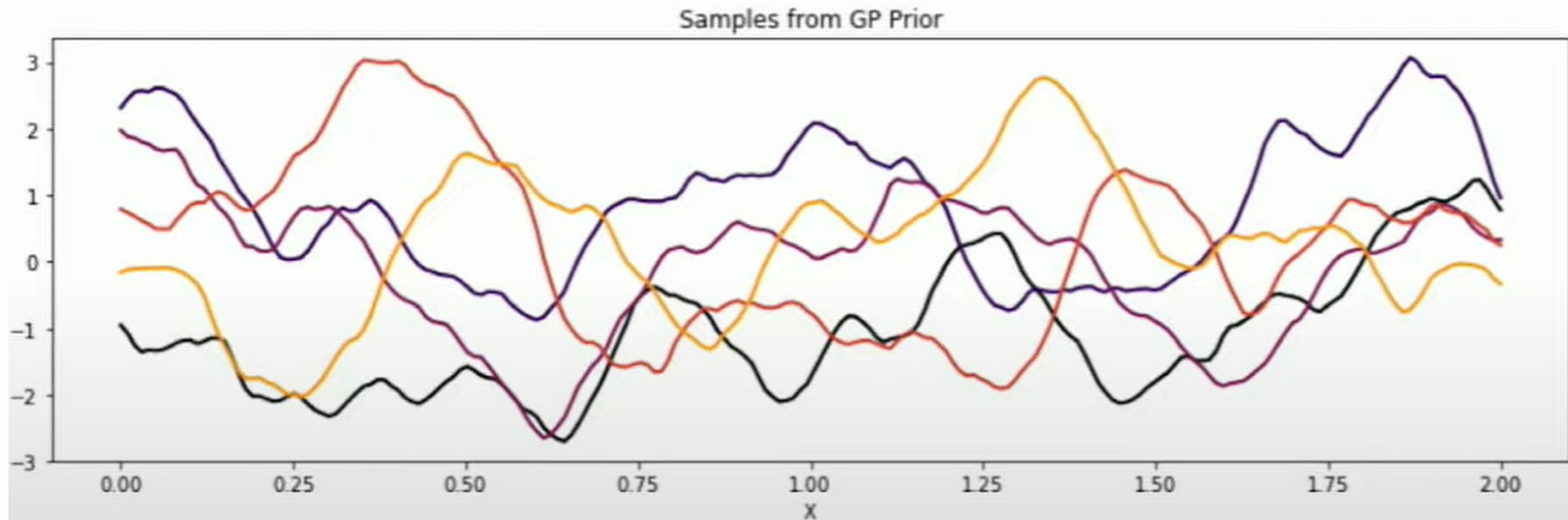


One parameter, length scale, that determines how far away you have to go away from two points to make them independent



Matern(3/2) is another popular covariance function

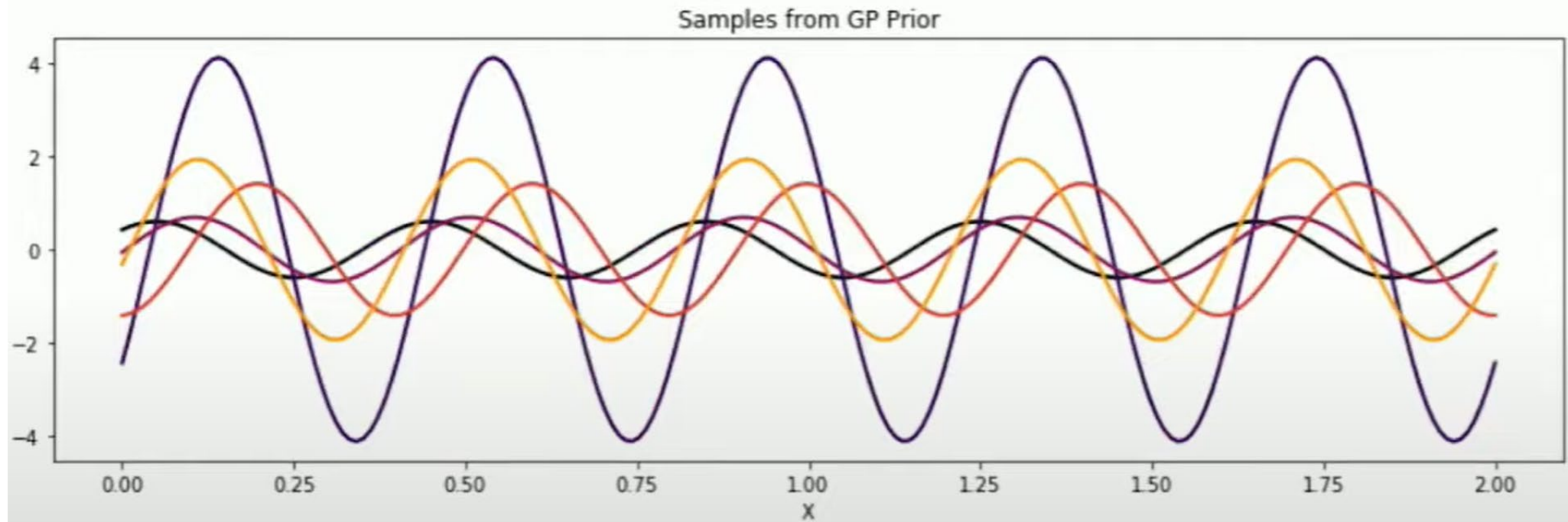
$$k(x, x') = \left(1 + \frac{\sqrt{3(x - x')^2}}{\ell} \right) \exp \left[-\frac{\sqrt{3(x - x')^2}}{\ell} \right]$$



Length scale parameter, plus a scaling term to scale roughness

Cosine is another popular covariance function

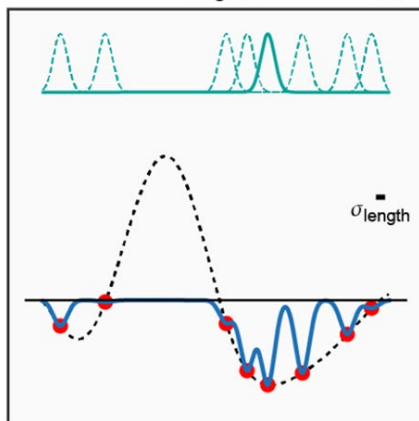
$$k(x, x') = \cos \left(\frac{\|x - x'\|}{\ell^2} \right)$$



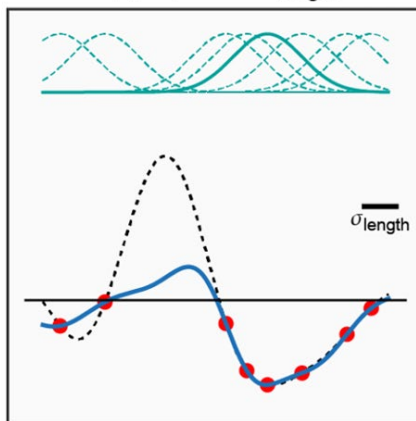
Getting the kernel hyperparameters right is important

Learning from function values

Too small σ_{length} (overfitting)

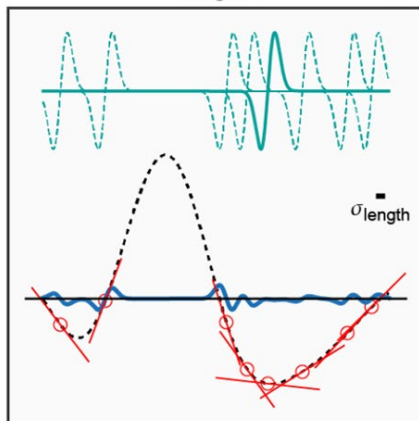


Appropriate σ_{length}

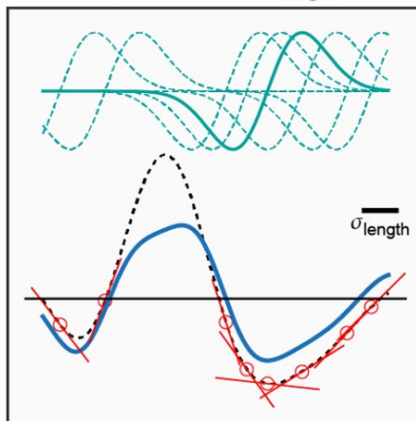


Learning from derivative values

Too small σ_{length} (overfitting)



Appropriate σ_{length}



A mean function is also necessary, but not as hard to pick

Mean function generates mean vectors

$$\mu \sim m(x|\phi)$$

Simplest is just zero!

$$m(x) = 0$$

Or constant

$$m(x) = C$$

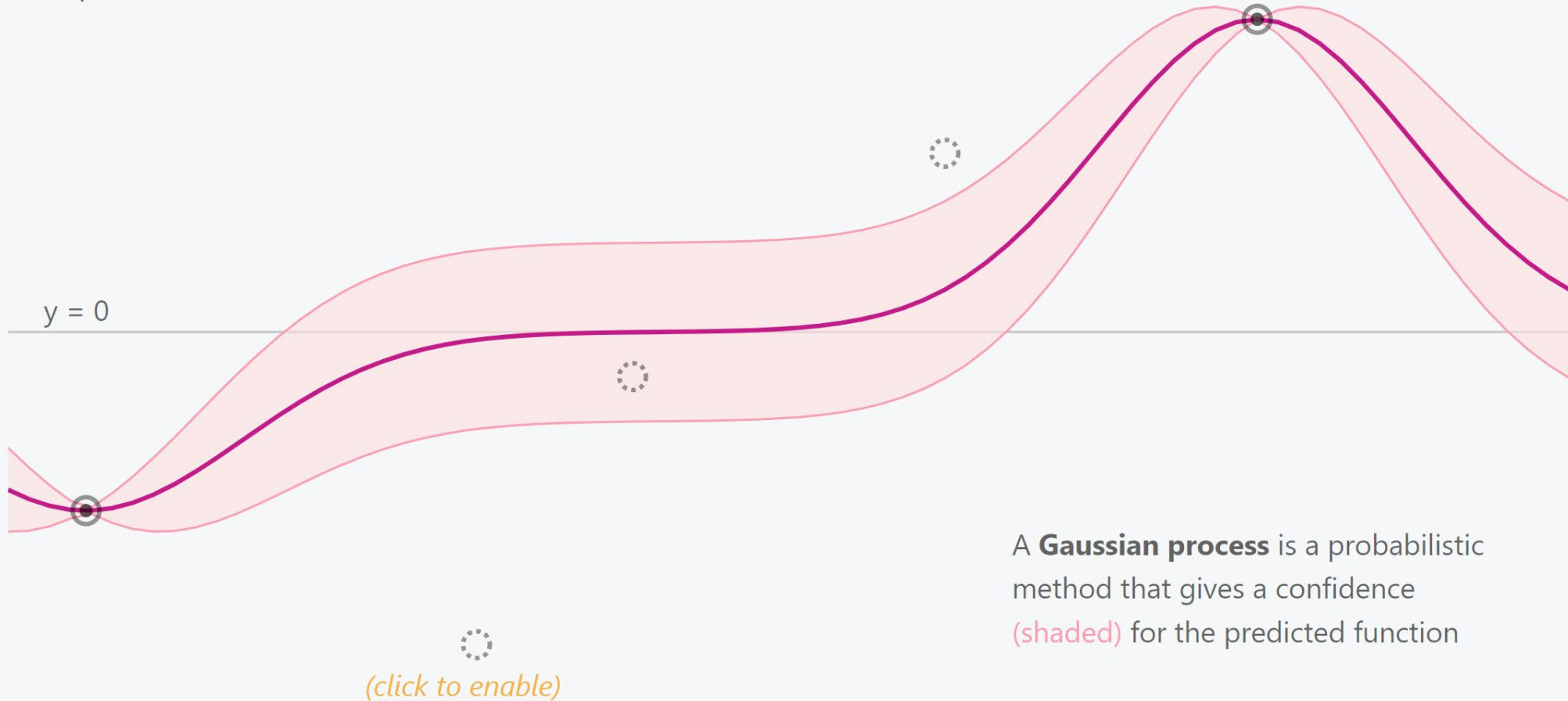
Or linear

$$m(x) = Ax + b$$

Posterior does not use the mean, it is just used as a guess for the prior because where you have no data, it trends towards the mean

In the absence of data, the model regresses towards the mean

Regression is used to find a function (line) that represents a set of data points as closely as possible





Use priors to calculate posteriors given some data observations

Recall Bayes' formula

$$p(\theta|y) \propto \prod_{i=1}^N p(y_i|\theta) p(\theta)$$

For Gaussian process, this becomes

$$\textit{Closed form GP} = \textit{Gaussian data} * \textit{GP prior}$$

When we fit the model, we are learning the kernel and mean hyperparameters



Predictions are done using Bayes theorem as well

Posterior predictive distribution

$$p(y^{new}|y) \propto \int p(y^{new}|\theta) p(\theta|y) d\theta$$

Likelihood is aleatoric uncertainty
Or noise in data itself

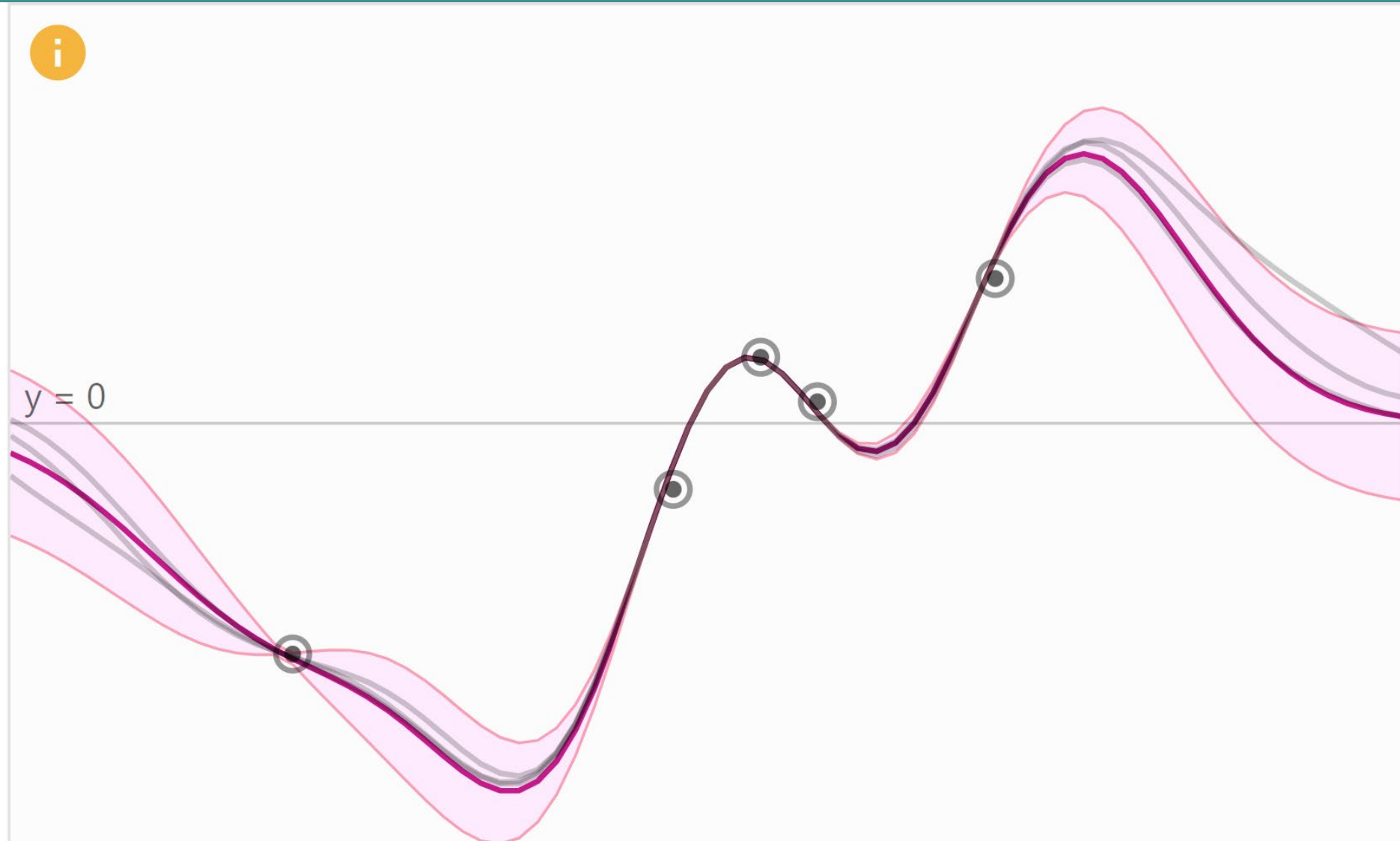
Posterior is epistemic uncertainty
Or our unknowns

Predictions are simply:

$$p(y^{new}|x^{new}, y, x) = N(\mu^{new}, \Sigma^{new})$$



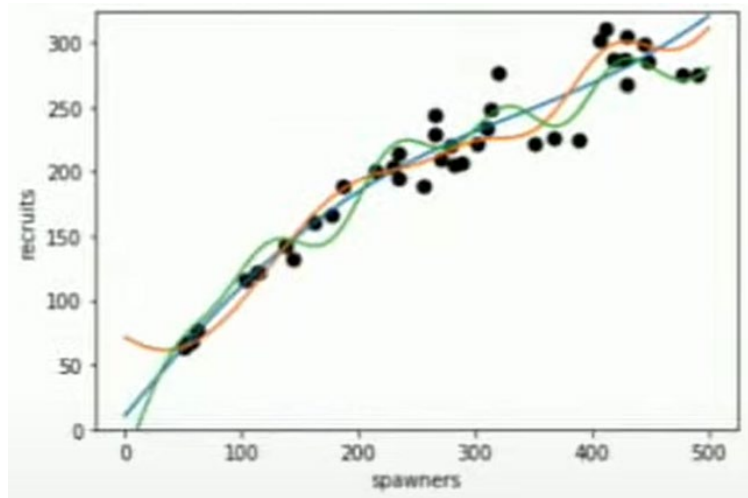
Confidence intervals and predictions come from distribution statistics μ, Σ



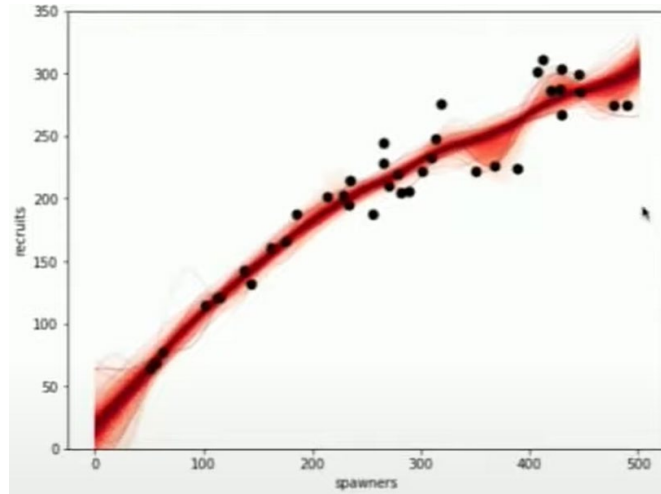


Confidence intervals and predictions come from distribution statistics μ, Σ

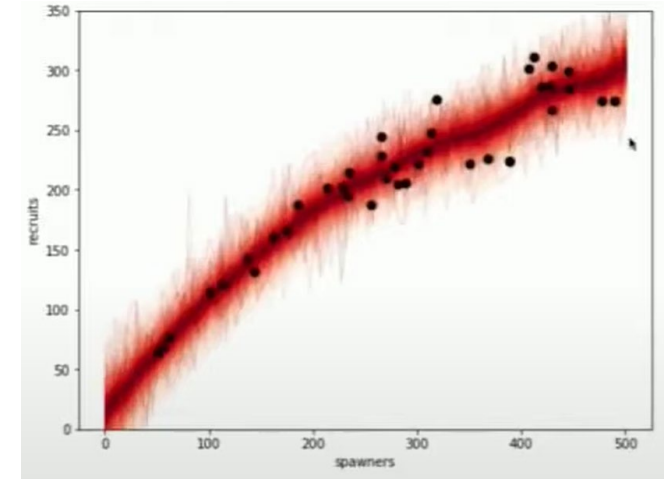
3 attempts at posteriors



100 attempts at posteriors



With prediction noise





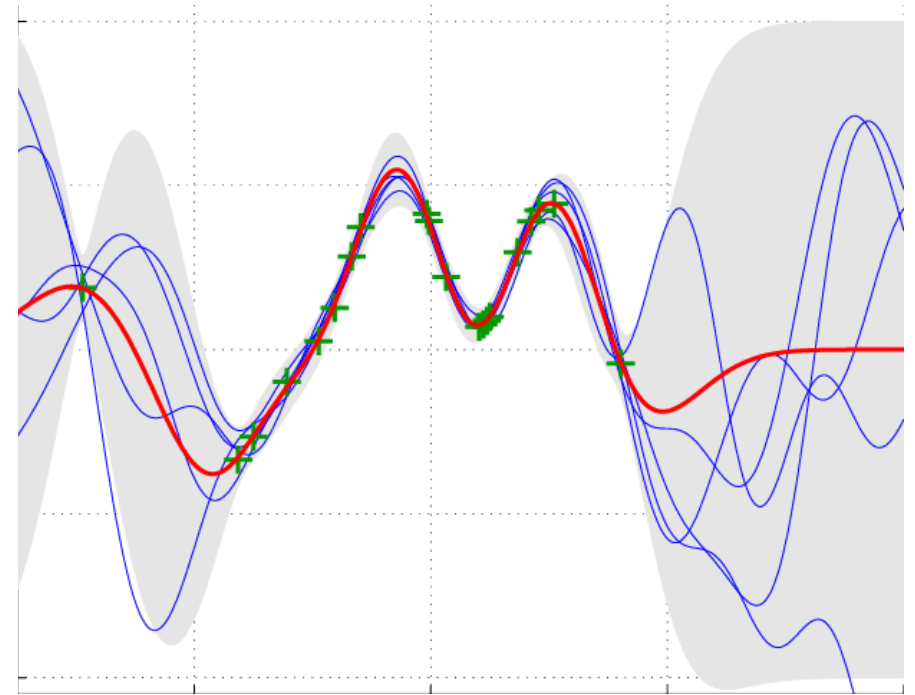
Gaussian Process does have some limitations!

Extrapolation does really poorly with Gaussian processes (depends on length scale of kernel)

They are not sparse (they use the whole samples/features information to perform the prediction)

They lose efficiency in high-dimensional spaces (features > a few dozen)

Picking and tuning kernel is tricky



Bayesian Optimization

